Citect for Windows
semAPI driver, User information and design

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Contents

1. USER INFORMATION 6

1.1 APPLICATION NOTES FOR DEVICE SEMAPI 6

1.1.1 Overview 6

1.1.1.1 General discussion 6

1.1.1.2 Supported physical interfaces 7

1.1.1.3 Backwards compatibility with existing Bailey serial driver 7

1.1.1.3.1 CitectSCADA tag address format 7

1.1.1.3.2 Function block support 7

1.1.1.3.3 Functions not supported in the semAPI driver 7

1.1.2 Setup guide 7

1.1.2.1 Installation of semAPI run-time 7

1.1.2.2 semAPI run-time under Windows NT® 8

1.1.2.2.1 Installing the semAPI device driver as a service 8

1.1.2.2.2 semAPI device driver started on request 8

1.1.2.3 semAPI run-time under Windows® 9x 8

1.1.2.4 semAPI run-time license entry 8

1.1.2.5 semAPI logical ICI configuration 8

1.1.3 Hints, Tips, and Frequently asked questions 10

1.1.4 Reference: Required components 10

1.1.5 Reference: Communications forms 10

1.1.6 Reference: Hardware configuration 11

1.1.6.1 Bailey INICI03A INFI-NET to Computer communication interface 11

1.1.6.2 Bailey INNIS01 Network Interface Slave Module 12

1.1.6.2.1 INNIS01 Dipswitch 1 (SW1) – Node Address 13

1.1.6.2.2 INNIS01 Dipswitch 2 (SW2) – Loop Address 13

1.1.6.2.3 INNIS01 Dipswitch 3 (SW3) – Loop Mode 13

1.1.6.2.4 INNIS01 Dipswitch 4 (SW4) – I/O expander bus (module) address & counters 13

1.1.6.3 Bailey INICT03A INFI-NET to computer transfer module 15

1.1.6.3.1 INICT03A Dipswitch SW1 – Baud Rate Switch 15

1.1.6.3.2 INICT03A Dipswitch SW2 – Diagnostics Switch 15

1.1.6.3.3 INICT03A Dipswitch SW3 – SCSI Switch 16

1.1.6.3.4 INICT03A Dipswitch SW4 – Option Switch 16

1.1.6.4 Bailey IMMPI01 Multi-function Processor Interface module 16

1.1.6.5 Bailey NTMP01 Multi-function Processor termination unit 17

1.1.6.6 Bailey NIMP01 Multi-function Processor termination module 18

1.1.7 Reference: Data types 19

1.2 DRIVER REFERENCE 23

1.2.1 Description 23

1.2.2 Driver generated error codes 23

1.2.2.1 Driver error ranges 23

1.2.2.2 Driver generated error numbers 23

Allowable Description 23

1.2.3 Parameters, options, and settings 30

1.2.4 Advanced 31

2. ANALYSIS 33

2.1 SEMAPI SYSTEM OVERVIEW 33

2.2 DRIVER DESIGN ANALYSIS 33

2.3 SEMAPI DEVELOPMENT ISSUES 34

3. DRIVER DESIGN 35

3.1 ADDRESSING 35

3.2 CITECTSCADA COMPILER MODIFICATIONS 35

3.3 FRONT-END/BACK-END DRIVER ARCHITECTURE 36

3.3.1 Driver memory tables 36
3.3.1.1 Exception report data table/Memory cache 37
3.3.1.2 Polled read data table 37
3.3.2 Driver initialisation 38
3.3.3 Driver data exception handling 39
3.3.4 Driver poll data read processing 40
3.3.5 Driver write handling 41
3.3.6 Driver read Processing 42
3.3.7 Multi-threading Protection 42
3.3.8 Driver flow examples 43
3.3.8.1 Unit initialisation for base variable 43
3.3.8.2 Unit initialisation for variable with sub-type 43
3.3.8.3 Exception report for communication interface index 5 43
3.3.8.4 CitectSCADA read request for real point 43
3.3.8.5 CitectSCADA read request for digital point 43

4. DRIVER IMPLEMENTATION 44

4.1 DEVELOPMENT RESOURCES 44
4.1.1 Documents 44
4.1.2 Software / Hardware 44

5. APPENDIX B – MODULE DEPENDANT STATUS' 45

5.1 NCTM01, IMAOM01, CTT01/02 45
5.2 NPIM01 45
5.3 NAMM01 45
5.4 NCOM02, IMCOM03/04, IMQRC01, CLC01/02/03/04, NLCM01/02/03 45
5.5 NLMM01 46
5.6 NLIM01/02, NBIM01, INBMI02 46
5.7 NAMM02/IMAMM03 46
5.8 INPTM01, NLSM01/INPCT01, CIC01 47
5.9 NMFC01/02, IMMFC03/04/05, IMMPC01, CBC01, CSB01, SLC01/21 47
5.10 IMLMM02 47
5.11 INBTM01, NLIM01/02 48
5.12 NGCM01/INPPT01 48
5.13 INIPT01 49
5.14 INIPT02 50
5.15 ICI MODULES: INICT01/03, IIMCP01/02 51
5.16 NSBM01 51
5.17 NSCM01 52
5.18 INIIT01/03 53
5.19 MFP MODULES: IMMFP01/02/03/03B, BRC-100 54
5.20 INNPM01 55
5.21 INIIT02 56
5.22 INCPM02/03 56
5.23 INSEM01 57
5.24 HARMONY PROCESSOR – CONTROLLER STATUS 58
5.25 HARMONY PROCESSOR – NPM STATUS 59

Table of Figures

FIGURE 1.1-1 CITECTSCADA SEMAPI DRIVER STRUCTURE 6
FIGURE 1.1-2 LOGICAL ICI CONFIGURATION APPLICATION – SERIAL 9

FIGURE 1.1-3 LOGICAL ICI CONFIGURATION APPLICATION – SCSI....9
FIGURE 1.1-4 HOST COMPUTER TO INFI-NET LOOP USING SCSI....11
FIGURE 1.1-5 HOST COMPUTER TO INFI-NET LOOP USING SERIAL....11
FIGURE 1.1-6 INICI03A MODULE WIRING DIAGRAM....12
FIGURE 1.1-7 INNIS01 MODULE BOARD LAYOUT....12
FIGURE 1.1-8 INICT03A MODULE BOARD LAYOUT....15
FIGURE 1.1-9 IMMPI01 MODULE BOARD LAYOUT....17
FIGURE 1.1-10 NTMP01 MODULE BOARD LAYOUT....17
FIGURE 1.1-11 NTMP01 RS232-C JUMPER SETTINGS....17
FIGURE 1.1-12 NTMP01 RS232-C DTE/DCE JUMPER SETTINGS....18
FIGURE 1.1-13 NIMP01 MODULE BOARD LAYOUT....18
FIGURE 1.1-14 NIMP01 RS232-C JUMPER SETTINGS....18
FIGURE 1.1-15 NIMP01 RS232-C DTE/DCE JUMPER SETTINGS....18
FIGURE 2.1-1 SEMAPI BLOCK DIAGRAM....33
FIGURE 3.3-1 DRIVER INITIALISATION FLOW DIAGRAM....38
FIGURE 3.3-2 DRIVER EXCEPTION HANDLING FLOW DIAGRAM....39
FIGURE 3.3-3 DRIVER POLL READ PROCESSING FLOW DIAGRAM....40
FIGURE 3.3-4 DRIVER CITECTSCADA WRITE-HANDLING FLOW DIAGRAM....41
FIGURE 3.3-5 DRIVER CITECTSCADA READ-HANDLING FLOW DIAGRAM....42

1. User information

1.1 Application notes for Device semAPI

	Detail
Manufacturer	Elsag Bailey (ABB)
Device name	semAPI/Bailey DCS
Communications method	RS-232C, SCSI

1.1.1 Overview

1.1.1.1 General discussion

Elsag Bailey is replacing several hardware interfaces available to enable communication between host computers and their DCS equipment with a single interface consisting of an Application Program Interface (API) and hardware communication interface (INICI03A). The API exposes standard functions to interface with the communication interface, independent on the underlying physical connection.
The DCS uses a variety of internal protocols to make up a Bailey network, including Module Bus, Control way, INFI-NET and Plant loop to communicate between modules, process control units (PCUs) and operator interface stations (OISs). The structure for the semAPI driver is shown in Figure 1.1-1.

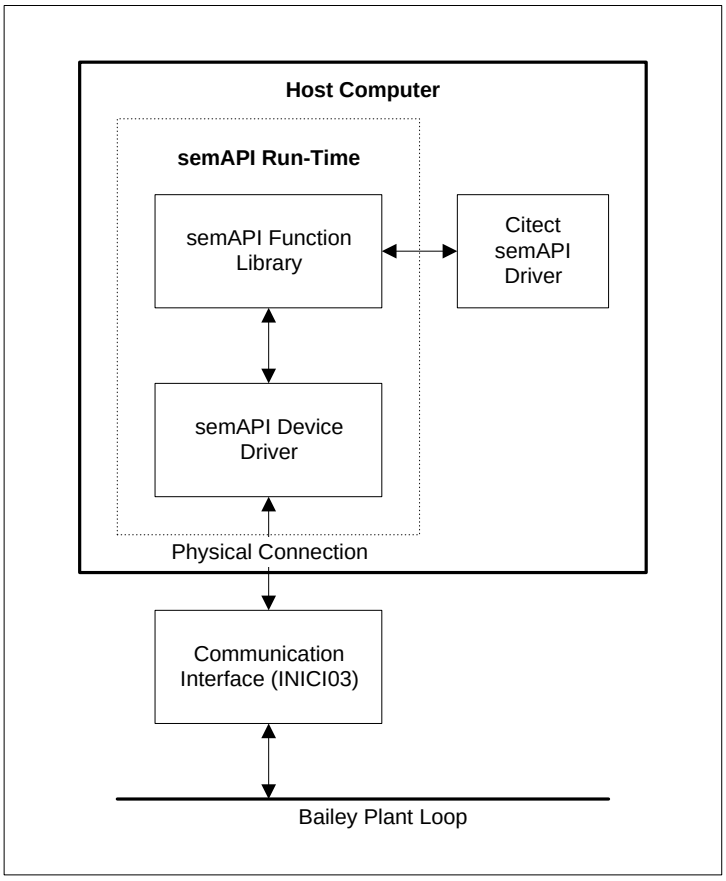


Figure 1.1-1 CitectSCADA semAPI driver structure

1.1.1.2 Supported physical interfaces

The semAPI driver will support communications to the Bailey network, through the communication interface, using any physical protocol supported by the semAPI run-time. Serial (RS-232C) and SCSI physical interfaces are currently supported by semAPI to communicate with the communication interface and it is anticipated that Elsag Bailey will also release support for a TCP/IP interface (harmonic coupler) in the future.

1.1.1.3 Backwards compatibility with existing Bailey serial driver

1.1.1.3.1 CitectSCADA tag address format

The addressing format used in the existing Bailey serial driver will be maintained within the new driver, where possible, both to provide backwards compatibility and to take advantage of the optimisations made within the CitectSCADA compiler for blocking (modified specifically for the existing Bailey Driver).
Note that the ability to use on WRCM point type to write to point types of RMC, DD and MSDD is no longer supported. Each point type has a separate write-block CitectSCADA data type.
The module status (MS) CitectSCADA data-type is not compatible with the existing Bailey driver as semAPI exposes significantly more status information.
The time set (TIME) CitectSCADA data-type is not compatible with the existing Bailey driver. This CitectSCADA data-type is now read/write. The read value gives the current Bailey system time/date as a CitectSCADA time/date variable. The write value allows the driver to synchronise the time to the time/date value specified in the write.

1.1.1.3.2 Function block support

The first phase of the semAPI driver will only provide support for the Bailey function blocks available in the existing Bailey driver. This is primarily due to the complexity of the Bailey addressing schema and the modifications to the CitectSCADA compiler for the existing Bailey driver.

1.1.1.3.3 Functions not supported in the semAPI driver

At current, the retail supervisory control version of the semAPI development kit does not provide support tune read/write or block read functionality. Due to this limitation, the semAPI driver cannot support the tune functionality (addresses of type TD, TI, TR) or block read functionality (BD, BI, BR).

1.1.2 Setup guide

1.1.2.1 Installation of semAPI run-time

To install semAPI Runtime Kit, obtain the semAPI Runtime Kit CD for Windows NT/9x (S-SB-API-X-4R00000000 Version 2.1 or later) from ABB/Elsag Bailey.
An installation guide is provided on the CD in pdf format.

Step 1

Run Setup.exe from the SEMAPI_RUNTIME directory on the semAPI Runtime Kit CD for Windows NT/9x® (S-SB-API-X-4R00000000 Version 2.1).

Step 2

Refer to the installation guide, summpage.pdf, for the semAPI Run-time Kit located on the semAPI Runtime Kit CD in the help directory SEMAPI_RUNTIME\document\i_sntir.

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1.1.2.2 semAPI run-time under Windows NT®

The are two methods for installing the semAPI device driver under this operating system.

1.1.2.2.1 Installing the semAPI device driver as a service

Step 1

The semAPI device driver may be installed as a service on the host computer. This is achieved by running the application DeviceAsService.exe from the semAPI run-time\exe\ directory. The format for the command is:

DeviceAsService.exe –i xxxxxxx

where xxxxxxx is the type of physical interface to use.

Examples:

DD_COM1 is for serial communications to serial port 1 (COM1)

DD_S004 is for SCSI communications to SCSI ID 4.

For the SCSI interface, the Windows NT machine must be booted with a functioning communication interface connected to the SCSI chain. (If the communication interface is not connected, Windows NT will not detect the communication interface and the service will fail to initialise).

Step 2

After the service has been installed, it must be started through the service control panel applet. The user may choose to configure the semAPI service to automatically start when Windows NT starts.

1.1.2.2.2 semAPI device driver started on request

The semAPI device driver will be automatically started when semAPI requires access to a physical communications interface. Note that the device will not shutdown automatically and must be manually terminated by the user before Windows 9x will complete shutdown.

1.1.2.3 semAPI run-time under Windows® 9x

Under the Windows 9x operating system, the same method described in 1.1.2.2.2.

1.1.2.4 semAPI run-time license entry

The user must install appropriate licensing through the license entry application supplied with the semAPI run-time software.

Consult the semAPI Run-Time Introduction and Installation instructions for further details.

1.1.2.5 semAPI logical ICI configuration

semAPI may support up to 20 different configurations for communications to host computer interface units. These configurations are edited through the application ICIConf.exe, supplied as part of the semAPI Run-Time software. Figure 1.1-2 shows the ICICONF configuration dialog.

The ICI configuration is comprised of the logical ICI number (1 – 20) and the type of physical connection (serial, SCSI). *The use of SCSI requires that the INICI03A be connected to the host computer's SCSI adaptor prior to Windows NT start-up. Selecting a SCSI physical connection is then done by selecting the SCSI ID from the Physical ICI device drop-down list (Sxxx where xxx is the SCSI ID).* Note that the license string for each configuration must match the 'User String' provided with the software licensing from Elsag Bailey.

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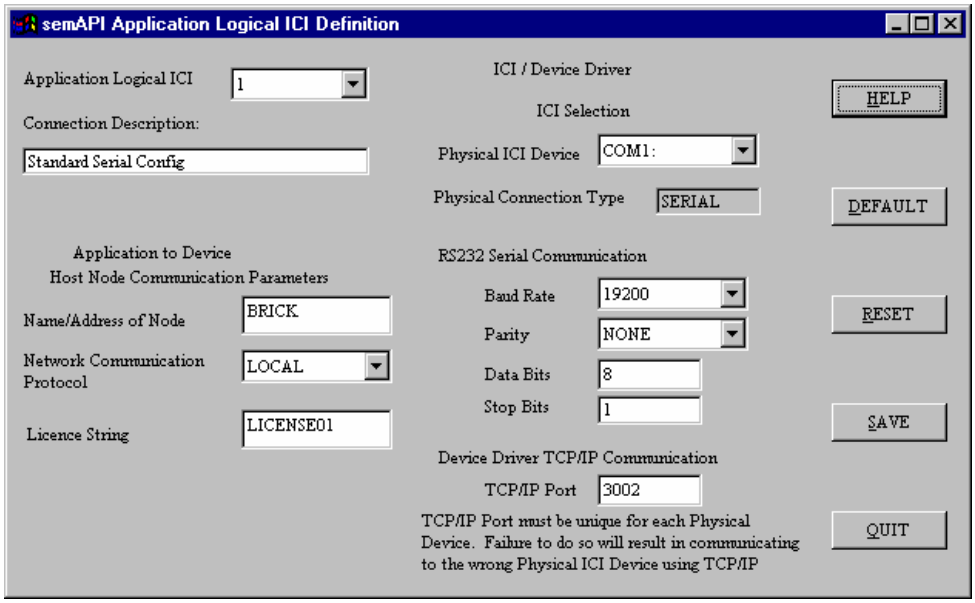


Figure 1.1-2 Logical ICI Configuration Application – Serial

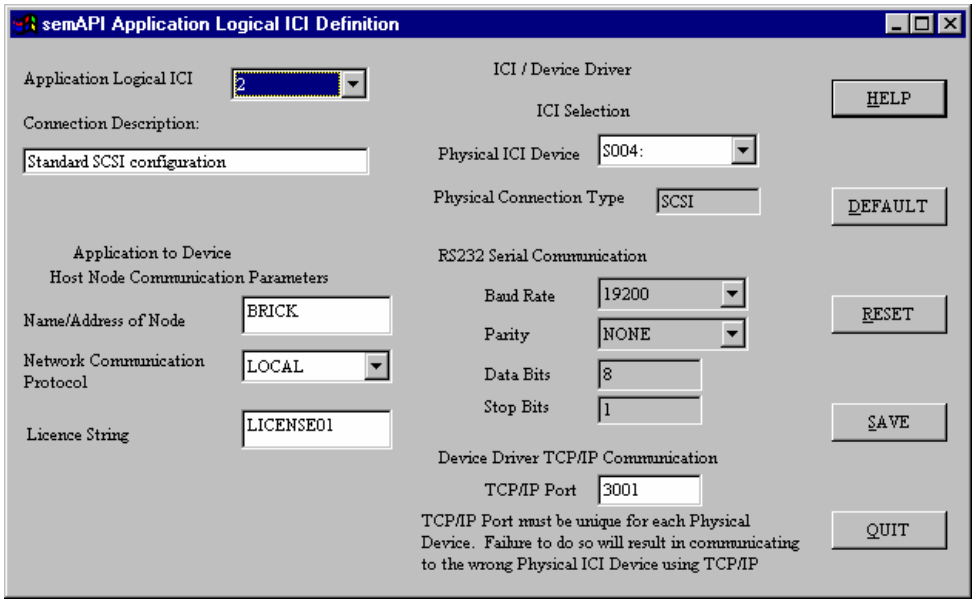


Figure 1.1-3 Logical ICI Configuration Application – SCSI

1.1.3 Hints, Tips, and Frequently asked questions

1.1.4 Reference: Required components

semAPI version 2.1 Run-Time software or later.
semAPI version 2.1 Run-Time Hardware protection key and software licensing or later.

1.1.5 Reference: Communications forms

Note that multiple ports may be allocated per board but only one unit is permitted per port.

Boards form

Field	Default	Allowable values
Board Name	This field is user defined, and is not used by the driver.	
Board Type	SEMAPI	
Address	0	
I/O Port		
Interrupt		
Special Opt		
Comment	This field is user defined and is not used by the driver.	

Ports form

Field	Default	Allowable values
Port Name	This field is user defined and is not used by the driver.	
Port number		
Board name	Refers to the board previously defined in ‘boards’ form.	
Baud rate		
Data bits		
Stop bits		
Parity		
Special Opt		
Comment	This field is user defined and is not used by the driver.	

I/O Devices form

Field	Default	Allowable values
Name	This field is user defined, and is not used by the driver.	
Number	Must be unique, but is not used by the driver.	
Address	Logical ICI configured in ICIConf.exe	1 – 20
Protocol	SEMAPI	
Port name	Refers to the port previously defined in ‘ports’ form.	
Comment	This field is user defined and is not used by the driver.	

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1.1.6 Reference: Hardware configuration

This section will describe the hardware and it's configuration, to permit communication to a Bailey INFI-NET network.

The communication interface, INICI03A, facilitates communication between a host computer and the Bailey INFI-NET loop. The INICI03A interface, referred to as the communication interface, consists of the following three modules:

- **INNIS01 Network Interface Slave (NIS) module:** this module allows the ICT module to communicate with other nodes within process control units (PCUs) on the INFI-NET loop
- **INICT03A INFI-NET to Computer Transfer (ICT) module:** this module handles communication between the host computer, through the Multi-function Processor Interface (MPI) module, and the Infi-net loop through the NIS.
- **IMMPI01 Multi-function Processor Interface (MPI) module:** this module provides the ICT module with communication ports. Ports provided include a SCSI port on the module's faceplate (see Figure 1.1-4), or a RS232-C serial port through a termination module (see Figure 1.1-5). Termination modules which provide this interface to the MPI include:
  - NIMP01/02 Multi-function Processor Termination Module;
  - NTMP01 Multi-function Processor Termination Unit.

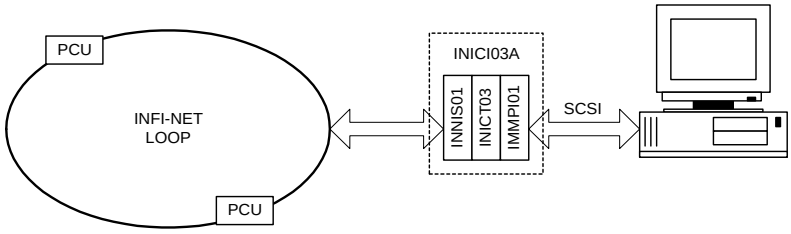


Figure 1.1-4 Host computer to INFI-NET loop using SCSI

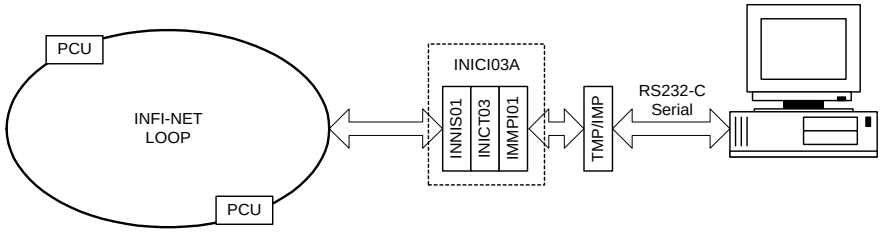


Figure 1.1-5 Host computer to INFI-NET loop using serial

This communication interface supports a maximum of 30,000 established indices (depending on point type and ICI memory). An index refers to a function block contained within a multi-function processor (MFP) or similar controller on the INFI-NET loop. The process of establishing a point creates a route from the communication interface to the MFP containing the INFI-NET address associated with the index (an INFI-NET address specifies the target MFPs loop, PCU and node address, then the function block within that MFP). When the state of an established function block changes, the MFP notifies the communication interface of the change through an exception report.

1.1.6.1 Bailey INICI03A INFI-NET to Computer communication interface

Figure 1.1-6 shows the connection topology for the modules making up the INICI03A interface. For detailed instructions on connection these modules refer to the Bailey communication module documentation for the INIC03A.

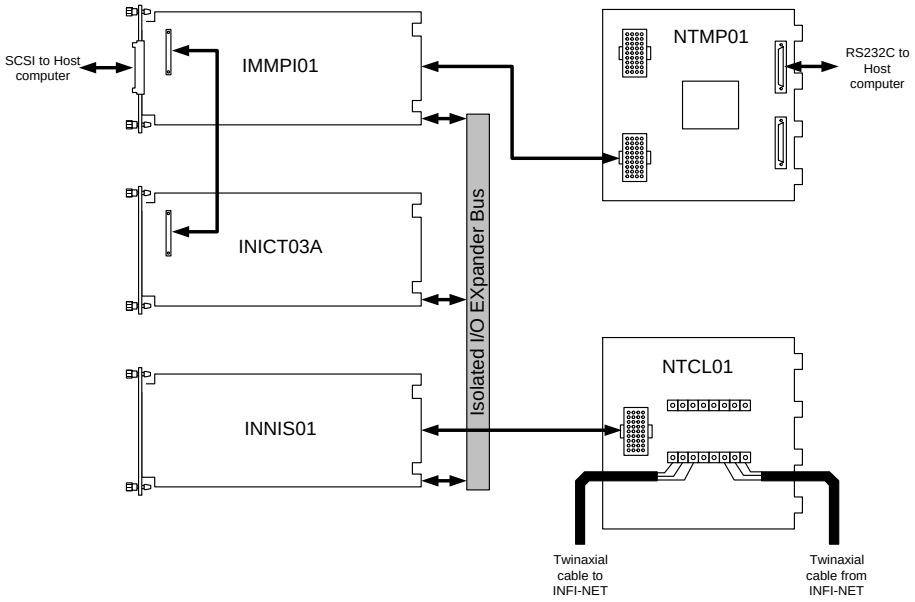


Figure 1.1-6 INICI03A Module Wiring Diagram

1.1.6.2 Bailey INNIS01 Network Interface Slave Module

This INNIS01 provides an interface for the INICT03A to the Bailey INFI-NET network. This section will detail the user configurable dipswitch settings. Further information on the INNIS01 can be found in the Bailey communication module documentation.

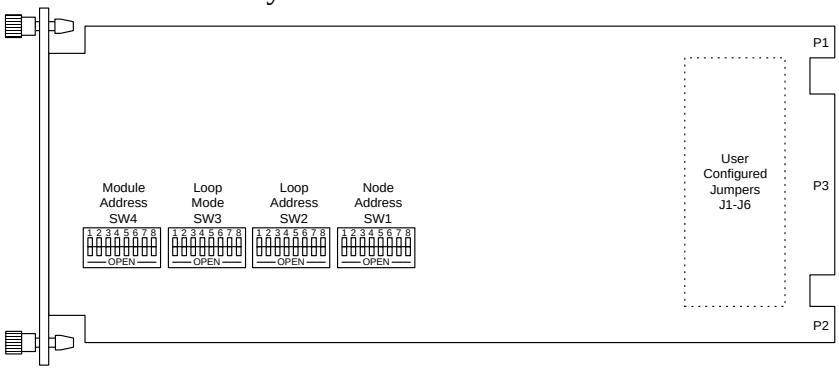


Figure 1.1-7 INNIS01 Module Board Layout

1.1.6.2.1 INNIS01 Dipswitch 1 (SW1) – Node Address

Specifies the module node address as a binary value with Pole 1 indicating the most significant bit.

Node Address	Dipswitch Pole (binary value)							
	Pole 1 (128)	**Pole 2 (64)**	**Pole 3 (32)**	**Pole 4 (16)**	**Pole 5 (8)**	**Pole 6 (4)**	**Pole 7 (2)**	**Pole 8 (1)**
1	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Open (1)
64	Closed (0)	Open (1)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)
250	Open (1)	Open (1)	Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)	Closed (0)

1.1.6.2.2 INNIS01 Dipswitch 2 (SW2) – Loop Address

Specifies the loop address, as a binary value, of the process control unit (PCU) interface (NIS) on the INFI-NET loop.

Loop Address	Dipswitch Pole (binary Value)							
	Pole 1 (128)	**Pole 2 (64)**	**Pole 3 (32)**	**Pole 4 (16)**	**Pole 5 (8)**	**Pole 6 (4)**	**Pole 7 (2)**	**Pole 8 (1)**
1	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Open (1)
64	Closed (0)	Open (1)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)
250	Open (1)	Open (1)	Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)	Closed (0)

1.1.6.2.3 INNIS01 Dipswitch 3 (SW3) – Loop Mode

Pole Position	Pole Open (1)	Pole Closed (0)
1	Unused	The NIS module is part of a PCU or INICI03A (INFI-NET to computer interface).
2	ROM check-summing enabled.	ROM check-summing disabled.
3	Test mode: no time-out for handshake failure.	Normal operating mode.
4	All loop messages return a busy negative acknowledge.	Normal operating mode.
5	Group A LEDs will toggle on & off if loop 1 is idle or shorted. Group B LEDs will toggle on & off If loop 2 is idle or shorted. Normal display otherwise.	LED Display as defined by Dipswitch SW4.
6	Diagnostic mode.	Normal operating mode.
7/8	Pole 7 Closed (0)/Pole 8 Closed (0) – 10 Mbaud INFI-NET mode.	
	Pole 7 Closed (0)/Pole 8 Open (1) – 2 Mbaud INFI-NET mode.	
	Pole 7 Open (1)/ Pole 8 Closed (0) – Unused.	
	Pole 7 Open (1)/ Pole 8 Open (1) – Unused.	

Note: black shaded pole positions indicate normal operating or preferred settings.

1.1.6.2.4 INNIS01 Dipswitch 4 (SW4) – I/O expander bus (module) address & counters

Poles 1 to 3 of SW4 determine the I/O expander module address as a binary value (see Section 1.1.6.2.4.1).

Poles 4 to 8 determine the function of LED counters on the front of the module (see Section 1.1.6.2.4.2).

1.1.6.2.4.1 INNIS01 Dipswitch 4 (SW4), Poles 1-3– Module Address

Module Address	Dipswitch Pole (binary value)		
	Pole 1 (128)	Pole 2 (64)	Pole 3 (32)
1	Closed (0)	Closed (0)	Closed (0)
64	Closed (0)	Open (1)	Open (1)
250	Open (1)	Open (1)	Open (1)

1.1.6.2.4.2 INNIS01 Dipswitch 4 (SW4), Poles 4-8 – LED function

Dipswitch Pole (Binary Value)					Description
Pole 4 (16)	Pole 5 (8)	Pole 6 (4)	Pole 7 (2)	Pole 8 (1)	
Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	# of timer interrupts.
Closed (0)	Open (1)	Closed (0)	Closed (0)	Open (1)	# of multicast messages received (excluding originated messages
Closed (0)	Open (1)	Closed (0)	Open (1)	Closed (0)	# of multicast destinations received.
Closed (0)	Open (1)	Closed (0)	Open (1)	Open (1)	# of time-sync messages received (excluding originated messages).
Closed (0)	Open (1)	Open (1)	Closed (0)	Closed (0)	# of broadcast messages received (excluding originated messages).
Closed (0)	Open (1)	Open (1)	Closed (0)	Open (1)	# of NIS poll messages received (excluding originated messages).
Closed (0)	Open (1)	Open (1)	Open (1)	Closed (0)	# of poll messages acknowledged by this node.
Closed (0)	Open (1)	Open (1)	Open (1)	Open (1)	# of poll messages busy negative acknowledged by this node.
Open (1)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	# of messages transmitted (total loop traffic).
Open (1)	Closed (0)	Closed (0)	Closed (0)	Open (1)	# of loop messages received & forwarded by this node.
Open (1)	Closed (0)	Closed (0)	Open (1)	Closed (0)	# of messages originated by this node (including retries).
Open (1)	Closed (0)	Closed (0)	Open (1)	Open (1)	# of message retries originated by this node.
Open (1)	Closed (0)	Open (1)	Closed (0)	Closed (0)	# of transmitted message watchdog expirations.
Open (1)	Closed (0)	Open (1)	Closed (0)	Open (1)	# of messages put into the receive buffer & retained.
Open (1)	Closed (0)	Open (1)	Open (1)	Closed (0)	# of bytes originated by this node (including retries).
Open (1)	Closed (0)	Open (1)	Open (1)	Open (1)	# of bytes received & forwarded by this node.
Open (1)	Open (1)	Closed (0)	Closed (0)	Closed (0)	# of I/O expander bus to NIS handshakes.
Open (1)	Open (1)	Closed (0)	Closed (0)	Open (1)	# of I/O expander bus to transmit buffer signals.
Open (1)	Open (1)	Closed (0)	Open (1)	Closed (0)	# of I/O expander bus process control unit (PCU) status requests.
Open (1)	Open (1)	Closed (0)	Open (1)	Open (1)	# of I/O expander bus NIS status requests.
Open (1)	Open (1)	Open (1)	Closed (0)	Closed (0)	# of I/O expander bus interrupts with invalid status.
Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)	# of transmit buffer realignments due to invalid contents.
Open (1)	Open (1)	Open (1)	Open (1)	Closed (0)	# of receive buffer realignments.
Open (1)	Open (1)	Open (1)	Open (1)	Open (1)	# of status buffer realignments.

Note: black shaded pole positions indicate normal operating settings.

1.1.6.3 Bailey INICT03A INFI-NET to computer transfer module

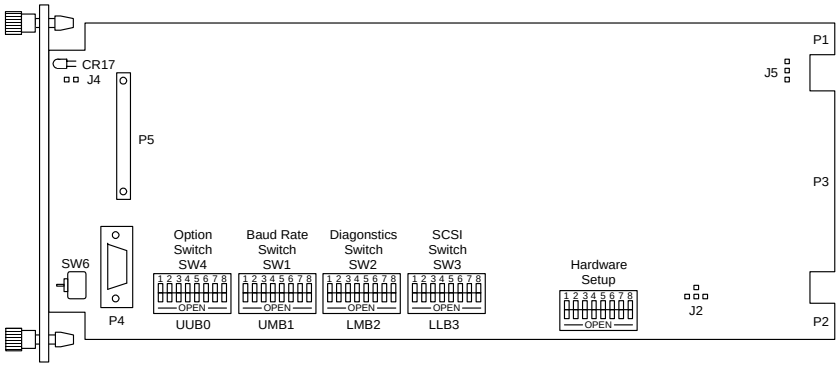


Figure 1.1-8 INICT03A Module Board Layout

1.1.6.3.1 INICT03A Dipswitch SW1 – Baud Rate Switch

Port Baud Rate	Terminal Port 1 Baud Rate Setting				Printer Port 2 Baud Rate Setting			
	Pole 1	Pole 2	Pole 3	Pole 4	Pole 5	Pole 6	Pole 7	Pole 8
50	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)	Closed (0)
75	Open (1)	Closed (0)	Closed (0)	Closed (0)	Open (1)	Closed (0)	Closed (0)	Closed (0)
110	Closed (0)	Open (1)	Closed (0)	Closed (0)	Closed (0)	Open (1)	Closed (0)	Closed (0)
134.5	Open (1)	Open (1)	Closed (0)	Closed (0)	Open (1)	Open (1)	Closed (0)	Closed (0)
150	Closed (0)	Closed (0)	Open (1)	Closed (0)	Closed (0)	Closed (0)	Open (1)	Closed (0)
300	Open (1)	Closed (0)	Open (1)	Closed (0)	Open (1)	Closed (0)	Open (1)	Closed (0)
600	Closed (0)	Open (1)	Open (1)	Closed (0)	Closed (0)	Open (1)	Open (1)	Closed (0)
1200	Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)	Open (1)	Open (1)	Closed (0)
1800	Closed (0)	Closed (0)	Closed (0)	Open (1)	Closed (0)	Closed (0)	Closed (0)	Open (1)
2000	Open (1)	Closed (0)	Closed (0)	Open (1)	Open (1)	Closed (0)	Closed (0)	Open (1)
2400	Closed (0)	Open (1)	Closed (0)	Open (1)	Closed (0)	Open (1)	Closed (0)	Open (1)
3600	Open (1)	Open (1)	Closed (0)	Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)
4800	Closed (0)	Closed (0)	Open (1)	Open (1)	Closed (0)	Closed (0)	Open (1)	Open (1)
7200	Open (1)	Closed (0)	Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)	Open (1)
9600	Closed (0)	Open (1)	Open (1)	Open (1)	Closed (0)	Open (1)	Open (1)	Open (1)
19200	Open (1)	Open (1)	Open (1)	Open (1)	Open (1)	Open (1)	Open (1)	Open (1)

Note: black shaded pole positions indicate recommended settings for semAPI.

1.1.6.3.2 INICT03A Dipswitch SW2 – Diagnostics Switch

Pole Position	Pole Open (1)	Pole Closed (0)
1	NIS handshake time-out disabled.	NIS handshake time-out enabled.
2	NIS diagnostic enabled.	NIS diagnostic disabled.
3	INFI-NET diagnostics enabled.	INFI-NET diagnostics disabled.
4	Hardware diagnostics enabled.	Hardware diagnostics disabled.
5	Not used.	
6	Not used.	
7	Not used.	
8	Not used.	

Note: black shaded pole positions indicate recommended settings for semAPI.

1.1.6.3.3 INICT03A Dipswitch SW3 – SCSI Switch

Pole Position	Pole Open (1)	Pole Closed (0)
1	SCSI Port enabled/serial port disabled.	SCSI port disabled/serial port enabled.
2/3/4	Closed (0)/Closed (0)/Closed (0) – SCSI Address 0	
	Closed (0)/Closed (0)/Open (1) – SCSI Address 1	
	Closed (0)/Open (1)/Closed (0) – SCSI Address 2	
	Closed (0)/Open (1)/Open (1) – SCSI Address 3	
	Open (1)/Closed (0)/Closed (0) – SCSI Address 4	
	Open (1)/Closed (0)/Open (1) – SCSI Address 5	
	Open (1)/Open (1)/Closed (0) – SCSI Address 6	
	Open (1)/Open (1)/Open (1) – SCSI Address 7	
5	SCSI parity checking enabled.	SCSI parity checking disabled.
6	Not used.	
7	Not used.	
8	Not used.	

Note: black shaded pole positions indicate recommended settings for semAPI.

1.1.6.3.4 INICT03A Dipswitch SW4 – Option Switch

Pole Position	Pole Open (1)	Pole Closed (0)
1	ROM check-summing disabled.	ROM check-summing enabled.
2/3[1]	Pole 2 Closed (0)/Pole 3 Closed (0) – Ports 0 & 1: 8 data bits, 1 stop bit, no parity (8N1)	
	Pole 2 Closed (0)/Pole 3 Open (1) – Ports 0 & 1: 8 data bits, 1 stop bit, even parity (8E1)	
	Pole 2 Open (1)/Pole 3 Closed (0) – Ports 0 & 1: 8 data bits, 1 stop bit, odd parity (8O1)	
	Pole 2 Open (1)/Pole 3 Open (1) – Ports 0 & 1: 8 data bits, 2 stop bits, no parity (8N2)	
4	Port 1 utility option. If this option is selected, port 1 data characteristics are automatically set to 8N1. Otherwise, port 1 data characteristics are set by dispswitch poles 2 and 3.	Port 1 option serial port to host.
5	Modem password protection enabled	Modem password protection disabled.
6	Port addressing mode enabled. When this option is enabled, the ICT module will expect all commands from the host to send the port address configured through the UTILITIES as the first character of each command.	Port addressing mode disabled.
7	Check-summing option enabled. When this option is enabled, the ICT module will expect all commands from the host to include a checksum byte in each reply.	Check-summing option disabled
8	Not used.	

Note: black shaded pole positions indicate normal operating or preferred settings.
[1] port 1 data characteristics are only applicable if pole 4 is closed.

1.1.6.4 Bailey IMMPI01 Multi-function Processor Interface module

The IMMPI01 provides the INICT03A with a SCSI port. Note that INICT03A SW3 Pole 1 must be open for the SCSI port to be enabled.

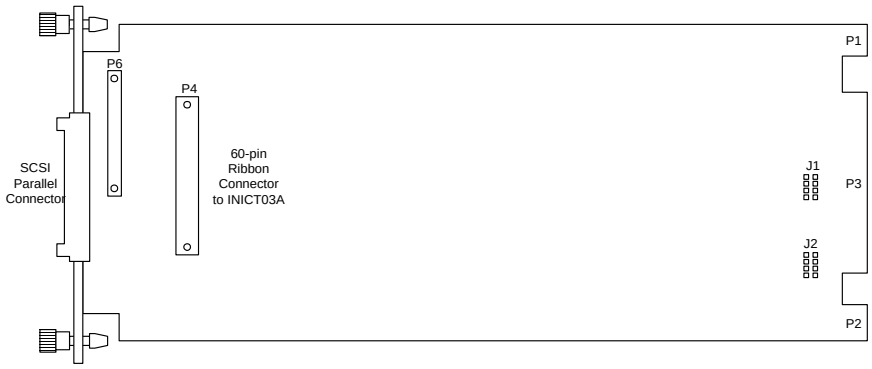


Figure 1.1-9 IMMPI01 Module Board Layout

1.1.6.5 Bailey NTMP01 Multi-function Processor termination unit

If using an NTMP01 termination unit, refer to Figure 1.1-10 and follow these instructions:

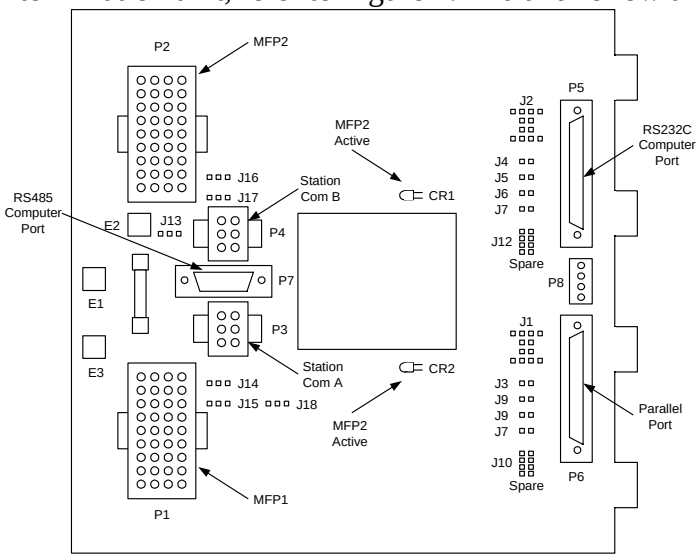


Figure 1.1-10 NTMP01 Module Board Layout

Step 1

If an RS-232-C connection is to be made to the INICI03A interface, set the computer port jumpers J4, J5, J6, and J7 as shown in Figure 1.1-11.

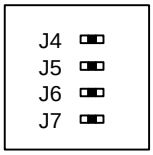


Figure 1.1-11 NTMP01 RS232-C Jumper Settings

Step 2

Determine the requirements of the application and set up the termination unit as DTE or DCE equipment by setting J2 accordingly as shown in Figure 1.1-12. The jumper settings shown are for RS232 communications with no hand-shaking.

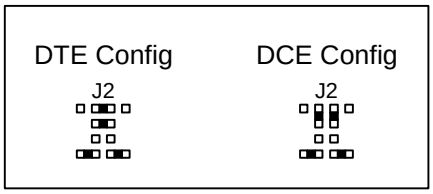


Figure 1.1-12 NTMP01 RS232-C DTE/DCE Jumper Settings

Step 3

Connect the appropriate end of the serial communication cable to P5 (computer port).

1.1.6.6 Bailey NIMP01 Multi-function Processor termination module

If using an NIMP01 termination module in an INFI 90 OPEN system, refer to Figure 1.1-13 and follow these instructions:

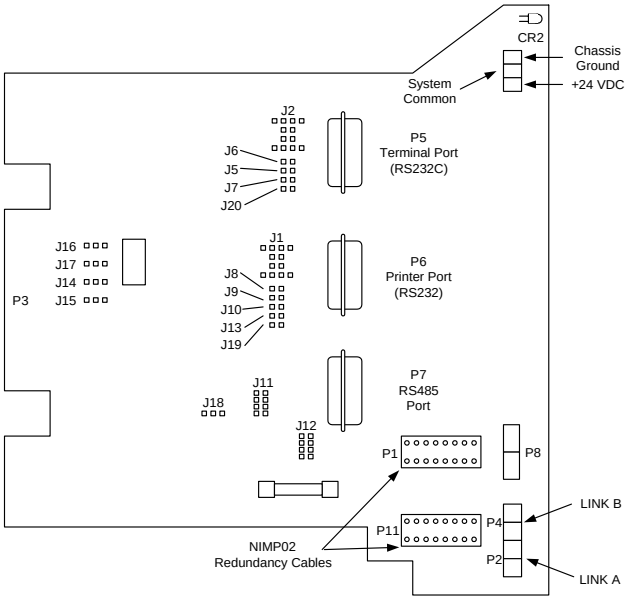


Figure 1.1-13 NIMP01 Module Board Layout

Step 1

If an RS-232-C connection is to be made to the INICI03A interface, set the computer port jumpers J6, J5, J7, and J20 as shown in Figure 1.1-14.

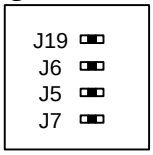


Figure 1.1-14 NIMP01 RS232-C Jumper Settings

Step 2

Determine the requirements of the application and set up the termination unit as DTE or DCE equipment by setting J2 accordingly as shown in iFigure 1.1-15. The jumper settings shown are for RS232 communications with no hardware hand-shaking.

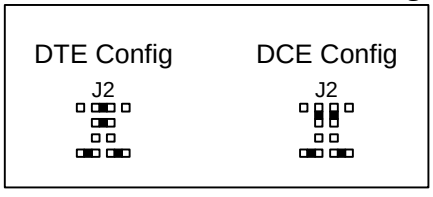


Figure 1.1-15 NIMP01 RS232-C DTE/DCE Jumper Settings

Step 3
Connect the appropriate end of the serial communication cable to P5 (computer port).

1.1.7 Reference: Data types

"xxxxx" is in the format <Index>, <Ring>, <Node>, <Module>, <Block>.

CitectSCADA Tag Format	Access Type	Data Type	Description
Axxxxx	Read Only	REAL	Analog block value
Axxxxx.BAD	Read Only	DIGITAL	Analog block bad quality flag
Axxxxx.HL	Read Only	DIGITAL	Analog block high limit alarm
Axxxxx.LL	Read Only	DIGITAL	Analog block low limit alarm
Axxxxx.HD	Read Only	DIGITAL	Analog block high deviation alarm
Axxxxx.LD	Read Only	DIGITAL	Analog block low deviation alarm
Axxxxx.RTG	Read Only	DIGITAL	Analog block one or more red tags
Axxxxx.SPT	Read Only	DIGITAL	Analog block set point tracking enabled
PVxxxxx	Read Only	REAL	Station block process variable value
PVxxxxx.BAD	Read Only	DIGITAL	Station block process variable bad quality flag
PVxxxxx.HL	Read Only	DIGITAL	Station block process variable high limit alarm
PVxxxxx.LL	Read Only	DIGITAL	Station block process variable low limit alarm
PVxxxxx.HD	Read Only	DIGITAL	Station block process variable high deviation alarm
PVxxxxx.LD	Read Only	DIGITAL	Station block process variable low deviation alarm
PVxxxxx.RTG	Read Only	DIGITAL	Station block process variable one or more red tags
PVxxxxx.SPT	Read Only	DIGITAL	Station block process variable set point tracking enabled
SPxxxxx	Read Only	REAL	Station block setpoint value
SPxxxxx.BAD	Read Only	DIGITAL	Station block setpoint bad quality flag
SPxxxxx.HL	Read Only	DIGITAL	Station block setpoint high limit alarm
SPxxxxx.LL	Read Only	DIGITAL	Station block setpoint low limit alarm
SPxxxxx.HD	Read Only	DIGITAL	Station block setpoint high deviation alarm
SPxxxxx.LD	Read Only	DIGITAL	Station block setpoint low deviation alarm
SPxxxxx.RTG	Read Only	DIGITAL	Station block setpoint one or more red tags
SPxxxxx.SPT	Read Only	DIGITAL	Station block setpoint set point tracking enabled
WSPxxxxx	Write Only	REAL	Station block setpoint write value
COxxxxx	Read Only	REAL	Station block control output value
COxxxxx.BAD	Read Only	DIGITAL	Station block control output bad quality flag
COxxxxx.HL	Read Only	DIGITAL	Station block control output high limit alarm
COxxxxx.LL	Read Only	DIGITAL	Station block control output low limit alarm
COxxxxx.HD	Read Only	DIGITAL	Station block control output high deviation alarm
COxxxxx.LD	Read Only	DIGITAL	Station block control output low deviation alarm
COxxxxx.RTG	Read Only	DIGITAL	Station block control output one or more red tags
COxxxxx.SPT	Read Only	DIGITAL	Station block control output set point tracking enabled
WCOxxxxx	Write Only	REAL	Station block control output write value
Rxxxxx	Read Only	REAL	Station block ratio value
Rxxxxx.BAD	Read Only	DIGITAL	Station block ratio bad quality flag
Rxxxxx.HL	Read Only	DIGITAL	Station block ratio high limit alarm
Rxxxxx.LL	Read Only	DIGITAL	Station block ratio low limit alarm
Rxxxxx.HD	Read Only	DIGITAL	Station block ratio high deviation alarm
Rxxxxx.LD	Read Only	DIGITAL	Station block ratio low deviation alarm

Rxxxxx.RTG	Read Only	DIGITAL	Station block ratio one or more red tags
Rxxxxx.SPT	Read Only	DIGITAL	Station block ratio set point tracking enabled
WRxxxxx	Write Only	REAL	Station block ratio write value
RMSCxxxxx	Read/Write	REAL	Remote Manual Set Constant block value status/write
RMSCxxxxx.BAD	Read Only	DIGITAL	Remote Manual Set Constant block bad quality flag
RMSCxxxxx.HL	Read Only	DIGITAL	Remote Manual Set Constant block high limit alarm
RMSCxxxxx.LL	Read Only	DIGITAL	Remote Manual Set Constant block low limit alarm
RMSCxxxxx.HD	Read Only	DIGITAL	Remote Manual Set Constant block high deviation alarm
RMSCxxxxx.LD	Read Only	DIGITAL	Remote Manual Set Constant block low deviation alarm
RMSCxxxxx.RTG	Read Only	DIGITAL	Remote Manual Set Constant block one or more red tags
RMSCxxxxx.SPT	Read Only	DIGITAL	Remote Manual Set Constant set point tracking enabled
Dxxxxx.BAD	Read Only	DIGITAL	Digital block bad quality flag
Dxxxxx.ALM	Read Only	DIGITAL	Digital block limit alarm
Dxxxxx.VAL	Read Only	DIGITAL	Digital block value
SSxxxxx.BYP	Read Only	DIGITAL	Station block status – Bypassed bad analog output active
SSxxxxx.MI	Read Only	DIGITAL	Station block status – Manual Interlock active
SSxxxxx.OT	Read Only	DIGITAL	Station block status – Output Tracking enabled
SSxxxxx.DSF	Read Only	DIGITAL	Station block status – Digital Station Failure
SSxxxxx.COK	Read Only	DIGITAL	Station block status – Computer OK signal received
SSxxxxx.LEV	Read Only	DIGITAL	Station block status – Station is at computer level
SSxxxxx.CRN	Read Only	DIGITAL	Station block status – Cascade/Ratio mode enabled
SSxxxxx.A	Read Only	DIGITAL	Station block status – Station is in automatic mode
SSxxxxx.BAD	Read Only	DIGITAL	Station block bad quality flag
SSxxxxx.HL	Read Only	DIGITAL	Station block high limit alarm
SSxxxxx.LL	Read Only	DIGITAL	Station block low limit alarm
SSxxxxx.HD	Read Only	DIGITAL	Station block high deviation alarm
SSxxxxx.LD	Read Only	DIGITAL	Station block low deviation alarm
SSxxxxx.RTG	Read Only	DIGITAL	Station block one or more red tags
SSxxxxx.SPT	Read Only	DIGITAL	Station block set point tracking enabled
SMxxxxx	Write Only	INT	Station block mode write: 0 = go to local-manual (station manual) 1 = go to local-auto (station auto) 2 = go to local-cascade/ratio (station c/r) 3 = go to computer-manual 4 = go to computer-auto 5 = go to computer-cascade/ratio 6 = go to local level 7 = go to computer level 8 = go to computer back-up state 9 = send computer OK signal 10 = go to station previous state
RCMxxxxx.Q	Read Only	DIGITAL	RCM block bad quality flag
RCMxxxxx.ALM	Read Only	DIGITAL	RCM block alarm
RCMxxxxx.TAG	Read Only	DIGITAL	RCM block one or more red tags
RCMxxxxx.OV	Read Only	DIGITAL	RCM block output value
RCMxxxxx.SI	Read Only	DIGITAL	RCM block logic set input active
RCMxxxxx.SP	Read Only	DIGITAL	RCM block set permissive active
RCMxxxxx.RI	Read Only	DIGITAL	RCM block logic reset input active
RCMxxxxx.OR	Read Only	DIGITAL	RCM block override active
RCMxxxxx.FB	Read Only	DIGITAL	RCM block feedback active

RCMxxxxx.SC	Read Only	DIGITAL	RCM block set command active
RCMxxxxx.RC	Read Only	DIGITAL	RCM block reset command active
WRCMxxxxx	Write Only	INT	RCM block mode write: 1 = RCM sustained reset 2 = RCM sustained set 5 = RCM pulsed reset 6 = RCM pulsed set
DDxxxxx.Q	Read Only	DIGITAL	DD block bad quality flag
DDxxxxx.ALM	Read Only	DIGITAL	DD block alarm
DDxxxxx.TAG	Read Only	DIGITAL	DD block one or mode red flags
DDxxxxx.V	Read Only	DIGITAL	DD block output value
DDxxxxx.F1	Read Only	DIGITAL	DD block input 1 feedback active
DDxxxxx.F2	Read Only	DIGITAL	DD block input 2 feedback active
DDxxxxx.FS	Read Only	DIGITAL	DD block input value bad quality flag
DDxxxxx.SO	Read Only	DIGITAL	DD block override active
DDxxxxx.M1	Read Only	DIGITAL	DD block mode bit 1 (DD Block in manual mode)
DDxxxxx.M0	Read Only	DIGITAL	DD block mode bit 0 (DD Block in Remote/Override mode)
WDDxxxxx	Write Only	INT	DD block mode write: 1 = reset control output 2 = set control output 4 = request manual mode 8 = request automatic mode
MSDDxxxxx.Q	Read Only	DIGITAL	MSDD block bad quality flag
MSDDxxxxx.ALM	Read Only	DIGITAL	MSDD block alarm
MSDDxxxxx.SO	Read Only	DIGITAL	MSDD block status override active
MSDDxxxxx.CO	Read Only	DIGITAL	MSDD block control override active
MSDDxxxxx.M	Read Only	DIGITAL	MSDD block auto mode
MSDDxxxxx.TAG	Read Only	DIGITAL	MSDD block one or mode red tags
MSDDxxxxx.V	Read Only	DIGITAL	MSDD block output value
MSDDxxxxx.F1	Read Only	DIGITAL	MSDD block input 1 feedback active
MSDDxxxxx.F2	Read Only	DIGITAL	MSDD block input 2 feedback active
MSDDxxxxx.F3	Read Only	DIGITAL	MSDD block input 3 feedback active
MSDDxxxxx.F4	Read Only	DIGITAL	MSDD block input 4 feedback active
MSDDxxxxx.GS1	Read Only	DIGITAL	MSDD block good state bit 1
MSDDxxxxx.GS0	Read Only	DIGITAL	MSDD block good state bit 0
MSDDxxxxx.RS1	Read Only	DIGITAL	MSDD block request state bit 1
MSDDxxxxx.RS0	Read Only	DIGITAL	MSDD block request state bit 0
WMSDDxxxxx	Write Only	INT	MSDD block mode write: 0 = request MSDD state 0 1 = request MSDD state 1 2 = request MSDD state 2 3 = request MSDD state 3 4 = request manual mode 8 = request automatic mode
RMCxxxxx.Q	Read Only	DIGITAL	RMC block bad quality flag
RMCxxxxx.ALM	Read Only	DIGITAL	RMC block alarm
RMCxxxxx.F1	Read Only	DIGITAL	RMC block input 1 feedback active
RMCxxxxx.F2	Read Only	DIGITAL	RMC block input 2 feedback active
RMCxxxxx.TAG	Read Only	DIGITAL	RMC block one or more red tags
RMCxxxxx.V	Read Only	DIGITAL	RMC block output value (1=running)
RMCxxxxx.BAD	Read Only	DIGITAL	RMC block bad start active

RMCxxxxx.FLT	Read Only	DIGITAL	RMC block is in fault
RMCxxxxx.SPM1	Read Only	DIGITAL	RMC block start permissive 1 state allowed
RMCxxxxx.SPM2	Read Only	DIGITAL	RMC block start permissive 2 state allowed
RMCxxxxx.E3	Read Only	DIGITAL	RMC block error bits E3=0, E2=0, E1=0, E0=0 – No errors in RMC E3=0, E2=0, E1=0, E0=1 – RMC is stopped E3=0, E2=0, E1=1, E0=0 – RMC Interlock 1 is zero E3=0, E2=0, E1=1, E0=1 – RMC Interlock 2 is zero E3=0, E2=1, E1=0, E0=0 – RMC Interlock 3 is zero E3=0, E2=1, E1=0, E0=1 – RMC Interlock 4 is zero E3=0, E2=1, E1=1, E0=0 – RMC Feedback 1 is zero E3=0, E2=1, E1=1, E0=1 – RMC Feedback 2 is zero E3=1, E2=0, E1=0, E0=0 – RMC Feedback 1 is one E3=1, E2=0, E1=0, E0=1 – RMC Feedback 2 is one
RMCxxxxx.E2	Read Only	DIGITAL	
RMCxxxxx.E1	Read Only	DIGITAL	
RMCxxxxx.E0	Read Only	DIGITAL	
WRMCxxxxx	Write Only	INT	RMC block mode write: 5 = RMC pulsed reset 6 = RMC pulsed set 129 = RMC acknowledge fault
MSxxxxx.MODE0	Read Only	DIGITAL	Module status MODE1=0 & MODE0=0 – Module in configure mode MODE 1=0 & MODE 0=1 – Module failed MODE 1=1 & MODE 0=0 – Module error MODE 1=1 & MODE 0=1 – Module in execute mode
MSxxxxx.MODE1	Read Only	DIGITAL	
MSxxxxx.ES	Read Only	DIGITAL	Errors exist for the module
MSxxxxx.ST3	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST4	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST5	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST6	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST7	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST8	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST9	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST10	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST11	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST12	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST13	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST14	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST15	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST16	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST17	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST18	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST19	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST20	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST21	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST22	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST23	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST24	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST25	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST26	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST27	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST28	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST29	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST30	Read Only	DIGITAL	Module specific meaning (see Appendix B)
MSxxxxx.ST31	Read Only	DIGITAL	Module specific meaning (see Appendix B)

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TIME	Read/Write	LONG	The Bailey system time/date, as a CitectSCADA time/date variable. When reading, if the time/date cannot be read, TIME will return –1. When writing, specify the new system time/date as a CitectSCADA time/date variable.

1.2 Driver reference

	Detail
Driver name	SEMAPI
Maximum array size[1]	256

1.2.1 Description

1.2.2 Driver generated error codes

1.2.2.1 Driver error ranges

Error Range	Allowable Description
10001 – 100FF	CitectSCADA semAPI driver errors
10100 – 101FF	semAPI ICI errors
10200 – 102FF	semAPI device driver errors
10300 – 103FF	semAPI message driver errors
10400 – 104FF	semAPI Sub-routine/function errors
10500 – 105FF	semAPI general errors

1.2.2.2 Driver generated error numbers

Error Number	Allowable Description
10001	Exception Processing error – processing will restart
10002	Cannot create driver thread – FATAL
10003	Invalid logical ICI
10004	Timeout while waiting for ICI status
10005	Cannot allocate memory
10006	Failed to set time, need to wait
10007	Unit shutdown requested by CitectSCADA
10008	Bad write arguments for write command
10101	Command queue to ICI, waiting for reply
10102	Improper Format for command
10103	Illegal Command issued
10104	Index already established in ICI
10105	Block Already established as another point
10106	Command is too long
10107	Bad reply from node interface
10108	Export used as import
10109	Second restart needed
1010A	Undefined index

[1] Equivalent to ‘Maximum Request Length’

1010B	Memory full
1010C	Host communication error
1010D	ICI internal module not responding
1010E	Import used as export
1010F	Timeout of plant loop response
10110	Number out of range
10111	Illegal Key used
10112	ICI requires restart command
10113	Module Status point used as import
10114	Message is active on the loop
10115	Import or export used as a module status
10116	Exception report specification lost
10117	No message queue for de-queue
10118	Reply to large
10119	Illegal station mode command
1011A	Illegal module number in command
1011B	Timeout between bytes in command
1011C	Index already established, by remote node
1011D	Point type incompatible with command
1011E	Watchdog timeout
1011F	Check-sum compare error
10120	Destination node is Off-Line
10121	Call-up command is required
10122	Computer interface internal error
10123	Computer interface is busy
10124	ICI has gone off-line
10125	Conflict with monitor mode
10126	Point type does not match ICI point type
10127	Destination loop is off-line
10128	Destination node is busy
10129	Destination loop is busy
1012A	Enhanced trend point is not established
1012B	UDXR Point was not established
1012C	Invalid Wallclock offset
10164	Undefined message type
10165	Module is busy, cannot reply
10166	Module mode conflicts with command
10167	Illegal message data
10168	Function block is not valid
10169	Function block is not configured
1016A	Function block as no readable parameters
1016B	Invalid function code specified for module
1016C	Function code and block number not compatible

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1016D	Insufficient memory in module to write block
1016E	Module is not responding
1016F	Specified input locked or tagged
10180	Waiting for module reply
101C8	No Security Device Present
101C9	Retry the ICI Command
101CA	Invalid Infi90 API Client detected
101CB	Wait and Retry the ICI Command
101CC	Invalid Security Device Present
10201	Another user owns ICI as exclusive
10202	ICI shared, exclusive not allowed
10203	Unknown connection type requested
10204	Application not connected.
10205	Message format error.
10206	Invalid message type.
1020A	error putting md header on
1020B	error filling trans struct
1020D	error communicating to ICI
1020E	check sum did not match reply
1020F	no response bytes received
10210	exclusive user already exists
10211	Invalid type connection
10212	token not available
10214	error sending to ICI
10215	number of users exceeds ick value
1021F	error initialising ICI NC
10220	error initialising MD NC
10221	reading communication file
10222	initialising device driver
10223	going to sleep
10224	checking the active list
10225	checking the response list
10226	routing message
10227	no nodes left on list
10228	no transaction structure
10229	error putting onto free list
1022A	error sending reply
1022B	error doing quick receive
1022C	Token not returned
1022D	Token owners not available
1022E	User Disconnect Error
1022F	Get Performance Data Error
10230	Quick Connect Error

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10231	Send Index Error
10232	Get Index Error
10233	Get Index List Error
10234	Clear Index Error
10235	Invalid Connection (Exclusive or Shared)
10236	No Wait connection found for quick
10237	unsolicited response to big for buffer
10238	cannot put unsolicited response on list
10239	cannot initialise a new user
1023A	cannot allocate memory for a new user
1023B	Error getting active transaction
1023C	Error getting user information
1023D	illegal index received
1023E	Client mismatch
1023F	Error deleting key message
10240	Error ALL keys in use
10241	ICK changed during normal operations
10242	Security Check to ICI failed
10243	Key has been cleared by the DD.
10244	Input buffer had bad header or message.
10245	User requested priority already allocated
10246	ICI reply was not properly encoded
10247	Command lost due to comm. Failure
10249	Failure to initialise SW license file
1024A	Failure to checkout SW license
1024B	Failure to checkin SW license
1024C	Failure to close (LicEnd) SW license file
10250	Autostart – Can't allocate transaction
10251	Autostart – Received unexpected response
10252	Autostart – Received failed response
10253	Autostart – Bad ICI type
10254	Autostart – Missing security device
10255	Autostart – Environment command failed
10301	Null pointer passed to decode by MD
10302	MD communication failed to initialise
10303	unable to get the requested token
10304	null decode funct received
10305	error checking inbuffer
10306	illegal retrieve command
10307	no msg on the pending list
10308	error checking messages
10309	putting message on list
1030A	error parsing info flags

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1030B	error getting node off list
1030C	null buffer struct received
1030D	no response from the DD
1030E	no message number returned
1030F	encoding header error
10310	error returning the token
10311	unable to get token for msg
10312	msg did not receive an ack
10313	msg had retry on ack list
10314	unable to send message to NC
10315	error putting msg on ack lis
10316	error mallocing message
10317	error connecting to ICI
10319	NC already connected
1031A	Invalid NC function call
1031B	NC software revision mismatch
1031C	NC no link (max clients connected)
1031D	Error message is not found
1031E	Error putting onto free list
1031F	Error setting server information
10320	Error getting buffer for message
10321	Invalid Byte Count in received msg
10322	Error checking message
10323	Error scheduling watchdog timer
10324	Error downloading ICI point table
10325	Error restarting the ICI
10326	Error putting ICI On or Offline
10327	Error stopping the DD
10328	Error copying the index file
10329	Error restoring indices to DD
1032A	Error deleting key
1032B	Product ID for ICK not Infi90 API
1032C	Number of Users is Invalid
1032D	Unsupported module type
1032E	Error sending ICK command to DD
1032F	First client still connecting to DD
10330	Error setting client priority in DD
10331	Error cancelling a keyed msg
10332	Server returned an unknown license type
10333	Indicates that SW failed and ICK is req.
10334	type connection is not a Bailey product
10335	wrong product ID when type connection is shared
10336	All Bailey products allowed, can not be specific product

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10337	User is already running error - can not connect again
10338	Failure during creation of the server
10339	Last client is shutting down DD
10401	Invalid logical ICI
10402	Invalid index
10403	Invalid number of points
10404	Invalid node type
10405	Invalid number of nodes
10406	Invalid index order
10407	Invalid auto disconnect
10408	Invalid Infi90 address
10409	Invalid point type
1040A	Invalid number of exceptions
1040B	null data pointer
1040C	Invalid ICI type
1040D	Invalid reply size
1040E	st_buffer is set to NULL
1040F	Bad reply code from ICI
10410	general Invalid arguments
10411	too many elements for group
10412	unable to allocate memory
10413	too many elements for list
10414	already connect to ICI
10415	connect to ICI failed
10416	user data area too small
10417	unable to lock ICI for restart
10418	Unable to read configuration
10419	Error establishing connect to ICI
1041A	Error establishing connect to ICI
1041B	Error initialising connection to ICI
1041C	Error initialising connection to ICI
1041D	Error disconnecting from ICI
1041E	Error disconnecting from ICI
1041F	Error disconnecting from server
10420	Error obtaining environment information
10421	Buffer length insufficient for reply
10422	Error establishing message system
10423	Error returning restart lock
10424	Error returning online lock
10425	Unable to lock ICI for online
10426	Error returning locks
10427	Error defining configuration (R) to server
10428	Error defining configuration (I) to server

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10429	Error failing over to backup ICI
1042A	Error on automatic restart of ICI
1042B	Unable to lock ICI for online command (p)
1042C	Unable to lock ICI for restart command (p)
1042D	Unable to lock ICI for manager command (p)
1042E	Fail-over is not valid for this configuration
1042F	Disconnect w/preceding connect
10430	Error disconnecting from ICI (Backup)
10431	Error disconnecting from ICI (Backup)
10432	Error disconnecting from server (Backup)
10433	Error communicating with backup
10434	Error updating user list (add)
10435	Error updating user list (delete)
10436	More exceptions returned that did not fit in the user data space
10438	Waiting on watchdog timer to timeout
10439	Error initialising SUB layer in connect
1043A	TAGNAME is not defined in database
1043B	Errors with SOME tags, check tag status codes
1043C	Illegal request sent to server
1043D	Illegal command sent to server
1043E	Invalid number of tags specified
1043F	Invalid number of exports specified
10440	Invalid option for DEFINE or UNDEFINE
10441	Invalid reply code returned from server
10442	Management is not enabled
10443	Unknown Problem Report returned by module
10444	Requested General Msg type is not supported
10445	Reply to General Msg was too small
10446	Requested Module Bus Msg type is not supported
10447	Reply to Module Bus Msg was too small
10501	error sending status to device driver
10502	error restarting ICI

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1.2.3 Parameters, options, and settings

Standard Parameters

Parameter	Default	Allowable values
Block (bytes)	256	
Delay (mS)	0	
MaxPending	10	
Polltime (mS)	0	This parameter should not be altered. The semAPI driver does not perform polling through the Driver Cpu function.
Timeout (mS)	0	
Retry	1	
WatchTime (Sec)	30	

Driver Specific Parameters

Parameter	Default	Allowable values
MaxReadExceptions	0	0 reads as many exceptions as possible. 2 to 255.
ExceptionScreening	0	0 to disable exception report screening. 1 to enable exception report screening.
SemAPIWatchTime	12 (30 secs)	This sets the hardware watchtime within both the semAPI device driver and the INICI03A. This value is multiplied by 2.5 to give a time in seconds (ie 1 → 2.5 secs). Valid values for this parameter are 2 to 255 representing the watchdog timer range of 5 secs to 637.5 secs.
MaxPendingPolls	10	Number of polled data reads that can be submitted concurrently to semAPI.
PolledDataPollTime (mS)	5000	Number of milliseconds between requests for a polled data read point.
ExceptionsToPollReads	10	Ratio of the number of exception processing loops to poll read loops. The higher the ratio, the more exceptions are processed in comparison to polled reads.
AutoRestartTimeOut (mS)	30000	The auto restart timeout in milliseconds. Increase this value if the driver requires more time to automatically re-establish indices in the communication interface. Symptoms of an insufficient timeout will be evident by the driver timing-out while waiting for an auto restart and performs a hard restart.
ThreadWaitTime (mS)	30000	Driver thread shutdown timeout. This parameter should only be changed at CitectSCADA Support's request.

ReadDcbDebug	0	Used for debugging CitectSCADA Read dcb data. When parameter is 1 and the driver is set to DEBUG_ALL, all read dcb data is dumped to the CitectSCADA Kernel. This parameter should only be changed at CitectSCADA Support's request.

1.2.4 Advanced

Driver generated statistics

Number	Label	Description
0	Read Proc. Loops	Total number of times the driver has 'looped' and checked to see if any exception reports or polled reads are available to be processed.
1	Read Proc. Errors	Total number of errors encountered while reading, causing the ICI to be restarted.
2	Read Proc. Warnings	Total number of warnings encountered while reading. (These warnings do not restart the ICI)
3	Rx Data Exceptions	The number of data read exception reports received from the communication interface and processed by the driver
4	Rx Spec Exceptions	The number of data specification exception reports received from the communication interface and processed by the driver.
5	Success Poll Reads	Number of successful polled reads performed.
6	Failed Poll Reads	Number of failed/unsuccessful polled reads performed.
7	Pending Polled Reads	Number of polled reads submitted to semAPI and data has not yet been returned.
8	Active Polled Reads	Number of polled reads that are currently waiting to be submitted to semAPI.
9	Total Sys Poll Reads	Number of polled reads in the driver cache. Over time, this will reflect the total number of polled reads in the CitectSCADA project.
6	Successful Writes	The count of successful writes issued by the driver to the semAPI/Bailey DCS.
7	Failed Writes	The count of unsuccessful writes issued by the driver to the semAPI/Bailey DCS.
8	Established Points	The number of indexes successfully established by the driver in the communication interface.
9	Establish Errors	The number of indexes which were unable to be established by the driver in the communication interface.
10	Hard restarts	Number of times the driver has gone 'offline' and performed a hard restart (ICI restart and establish indices).

11	Auto restarts	Number of times the driver has gone 'offline' and has been restarted using the semAPI ICI management functionality (which automatically downloads index table to communication interface).

Debug messages explained

Debug messages will be displayed in the CitectSCADA Kernel if the driver debug level is set to DEBUG_ALL. If the debug level is set to DEBUG_WRITE, only write debug messages will be displayed. These debug messages display actions that the semAPI driver is performing or has performed, describing the action and an error number if an error has occurred.
An example of the debug messages generated for a successful startup are shown below:

```
(ICI x) Unit Init {}
(ICI x) Init Unit {Starting Timer}
(ICI x) Init Thread {Performing Hard restart}
(ICI x) Init Thread {Connecting to ICI...}
(ICI x) Init Thread {Successfully connected}
(ICI x) ICI Init {Attempting to hard restart ICI...}
(ICI x) ICI Init {Restart successful. Establishing indices...}
(ICI x) ICI Init {Index establish complete...}
(ICI x) Init Thread {Resuming exception processing}
(ICI x) Init Thread {Init successful}
```

An example of an unsuccessful startup (with no physical connection to the ICI) is shown below:

```
(ICI x) Unit Init {}
(ICI x) Init Unit {Starting Timer}
(ICI x) Init Thread {Performing Hard restart}
(ICI x) Init Thread {Connecting to ICI...}
(ICI x) Init Thread {Failed to connect to ICI, Error = 0x10415}
(ICI x) Init Thread {Init Failed, Error = 0x10415}
```

Write debug messages will be displayed if the driver debug level is DEBUG_WRITE or DEBUG_ALL. The message will contain the address of the point, the value that the driver has attempted to write and an error number, if applicable.

Example debug messages for writing to various blocks are shown below:

```
(ICI x) Write {MSDD2004,1,4,4,303=4, Error = y}
(ICI x) Write {WSP2003,1,4,4,201=15.50000, Error = y}
```

2. Analysis

2.1 semAPI system overview

The semAPI provides an interface to the Bailey INFI90 Network by communicating with an INICI03A communication interface. semAPI function calls pass through a device driver which provides hardware/physical communications abstraction. Through this method, no changes are required to the calling application if the communication media changes.
A block diagram of the semAPI driver is shown in Figure 2.1-1. This includes the structure for semAPI client/server configuration, where the semAPI client may reside on a computer other than the computer physically connected to the INICI03A. When the client and server are located on a single machine, the same structure applies, however network communications are conducted on the same communications layer (ie network packets never reach the physical network layer).
The CitectSCADA semAPI driver has not been tested using physically separated semAPI clients and servers.

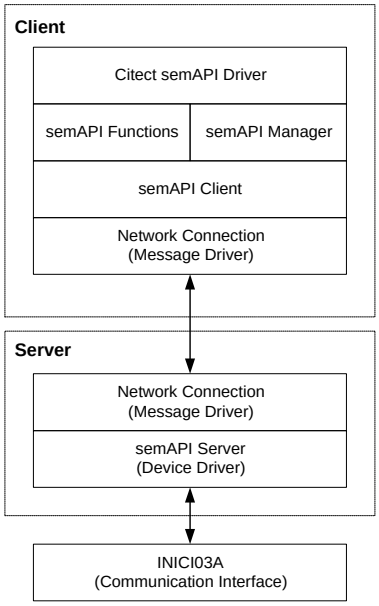


Figure 2.1-1 semAPI Block Diagram

The semAPI manager handles the automatic recovery of communications to the INICI03A, including rebuilding the index table, by an application enabling semAPI management functionality (see Section **Error! Reference source not found.**).

2.2 Driver design analysis

The preliminary driver design involved the use of three memory tables. The first table providing look-up functionality for all tags in the CitectSCADA project(s). The table would then provide a link to either the exception report table or the polled read table depending on the tag's type. This design strategy was abandoned due to the memory overhead and the excessive dynamic memory allocation.

The design strategy chosen uses an exception report table and polled read table. This approach is more suitable due to simplicity and less requirements for dynamic memory management.

The exception report table is an arbitrary size (30,000 records) relating to the maximum number of indices that can be stored in the INICI03A. Each record in the table stores the value of the last exception report returned to CitectSCADA from semAPI. Note that an exception report is generated for each established point when the INICI03A is set to on-line mode.

The polled read table is dynamically sized during driver run-time. When a CitectSCADA requests a polled read address, it is added to this table if it does not already exist. Note that elements in this table will only be released when the unit is stopped at driver shutdown or when the unit is disabled using the Cicode function IODeviceControl). Any CitectSCADA requests for polled data are immediately serviced with current cached data. At this stage, only module status and system time/date are supported as polled reads. Future enhancements to the semAPI development kit may permit access to block specs (tune blocks) and block reads. These results of these reads will be stored as part of the polled read table.

Module status information is partially exception reported (errors exists and mode status') with the majority of status information returned by the semAPI function s_demand_module_status (see **Error! Reference source not found.**). At the completion of a demand module status polled read, the status information is copied to the structure for the module status index in the exception report table. System time/date is determined through the semAPI function s_read_system_date_time_w (see **Error! Reference source not found.**), with the result stored as a member of the UNIT_DB structure.

2.3 semAPI development issues

This section will describe issues that have been encountered during the driver implementation and prototyping.

The semAPI function s_read_ici_status only returns the status of the ICI at the time of the last semAPI function that required communications to the ICI. This means that any functions that only access up to the semAPI device driver layer, such as s_get_data_except, do not update the ICI status to the actual status. The implemented work around for has been to call s_envirnoment_w prior to the s_read_ici_status call.

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3. Driver Design

3.1 Addressing

Most of the semAPI commands require the address of a specific block location within the DCS system. Bailey uses a proprietary addressing system, which does not map well into the addressing scheme used by CitectSCADA. The system refers to a specific block within the DCS using the following structure (see INFI90ADR structure in Section 4.7):

Name	Range	Description
Ring	0 – 7	The loop that the PCU is on
Node	0 – 255	The PCU number
Module	0 – 31	The module within the PCU
Block	0 – 16383	The block within the module.

For exception reporting purposes, routes, which map a block within the DCS into the communication interface, are referred to by:

Name	Range	Description
Index	0 – 30000	The point number in the communication interface.

To facilitate this in the CitectSCADA driver, the DATAPOINT structure has been mapped into the RDBDATA structure. Therefore UnitAddress and UnitType fields are assigned as follows at compile time.

UnitAddress Bits	Description
Bit 31	Write flag
Bit 30	Read flag
Bits 29 – 24	Point type (PV, SP, CO, RI, A, SS, D, SM, MS, RCM, RMSC etc)
Bits 23 – 16	This byte will be put into UnitAddress when 0 is entered as an index (applicable for tune or block reads)
Bits 15 – 0	Index

UnitType Bits	Description
Bits 29 – 16	Block
Bits 15 – 11	Module
Bits 10 – 3	Node
Bits 2 – 0	Ring

3.2 CitectSCADA compiler modifications

Modifications were made to the CitectSCADA compiler to permit the blocking of Client/Server communications for systems using the Bailey Driver. These modifications are:

An additional rdb (_driver.rdb) contains the raw information as would have been generated for _variabl.rdb if the standard compiler were used (using the driver address format specifications).
_variabl.rdb is compiled specially with the following rules applied:

- All variables, regardless of the raw data type have UnitType = 0 (this is to allow blocking of Client/Server requests);
- The CitectSCADA unit address of variables is generated by the compiler using the following rules:
  a) for points with a raw data type of RDT_REAL:
     CitectSCADA Unit Address = Index
  b) for points with a raw data type of RDT_DIGITAL:
     CitectSCADA Unit Address = Index << 5 + SubType Index

The driver uses the _driver.rdb to obtain the actual INFI90 address during the initialisation phase. This address is required when establishing points within the communication interface.

During driver operations, CitectSCADA unit addresses are converted back to communication interface indices and sub-type indices for look-up in the memory cache or use in semAPI write functions.

Within the driver, unit addresses are converted to a communication interface index and sub-type index using the following rules:

- for points with a CitectSCADA raw data type of RDT_REAL:
  Communication Interface Index = CitectSCADA Unit Address
  Sub-Type Index = -1
- for points with a CitectSCADA raw data type of RDT_DIGITAL:
  Communication Interface Index = CitectSCADA Unit Address >> 5
  Sub-Type Index = CitectSCADA Unit Address & 0x1F

3.3 Front-end/Back-end driver architecture

The semAPI driver will be a front-end/back-end type driver, with CitectSCADA read requests serviced from a memory cache of block values within the Bailey DCS system. A driver back-end thread is used to update the memory cache by servicing data exception reports generated when established blocks vary within the Bailey DCS network. CitectSCADA write requests will be queued and serviced by another back-end thread, which will perform each queued write operation using the semAPI function, s_output_control_point_w.

3.3.1 Driver memory tables

The following memory tables form part of the unit database structure for the driver.

3.3.1.1 Exception report data table/Memory cache

Location: Back-end.

Description: The number of element in this table is arbitrarily set to the maximum number of indices supported by the communication interface. Any unused communication interface indices will leave 'holes' within this table.

This table is indexed according to the point index within the communication interface; ie element 1 corresponds to point index 1 in the communication interface.

When an exception report is received by the driver back-end, the received exception data will be copied to the corresponding index elements' Data Structure field and the Time-stamp field updated to that of the exception report's time-stamp. Note that polled reads for module status will update the exception report data for the index associated with the module status.

Table Structure:

Data	The data for the point. (Bailey data structure VALUE_TABLE).
Time-stamp	The time-stamp of the last received exception report for point. Bailey data structure ICI_TIMESTAMP.
Used	A flag indicating whether this index is used by the CitectSCADA system.
Established	A flag indicating whether this index has been established successfully within the communication interface.

3.3.1.2 Polled read data table

Location: Back-end.

Description: This table is built dynamically by the driver.

When CitectSCADA requests a point that must be polled (rather than exception reported), an entry is made in this table if it doesn't already exist.

If a request is not outstanding (ie. quick message ID structure, ID member is 0) and the current tick time exceeds the next update time then a request for the point will be placed on the poll request queue which is periodically handled by the back-end driver.

Table Structure:

Data	The data for the polled point. A union structure of the Bailey data structures VALUE_TABLE and STATUS_TABLE.
CitectSCADA UnitAddress	The unit address for the read request.
NextUpdateTime	The time when which a new request for the poll read element can be made.
Quick message ID	The semAPI quick message ID structure (see **Error! Reference source not found.**) for an outstanding request. This structure is set when a semAPI function to request the read is invoked and is reset when the quick response is retrieved.
Polled Data	To be determined.

3.3.2 Driver initialisation

The driver initialisation flow diagram is shown in Figure 3.3-1. The initialisation task needs to be executed in a separate thread to the main driver, as the semAPI functions used to initialise the ICI are blocking. This thread is started when CitectSCADA issues an Init Unit Dcb to the semAPI driver. A timer is used to ensure that the driver does not hold the Unit Init Dcb longer than the unit watch time, replying to Dcbs with the driver error, *'unit offline cannot talk'*, until the thread has completed. The data exception handling and write-handling threads are started only if the driver is able to successfully initialise the ICI.

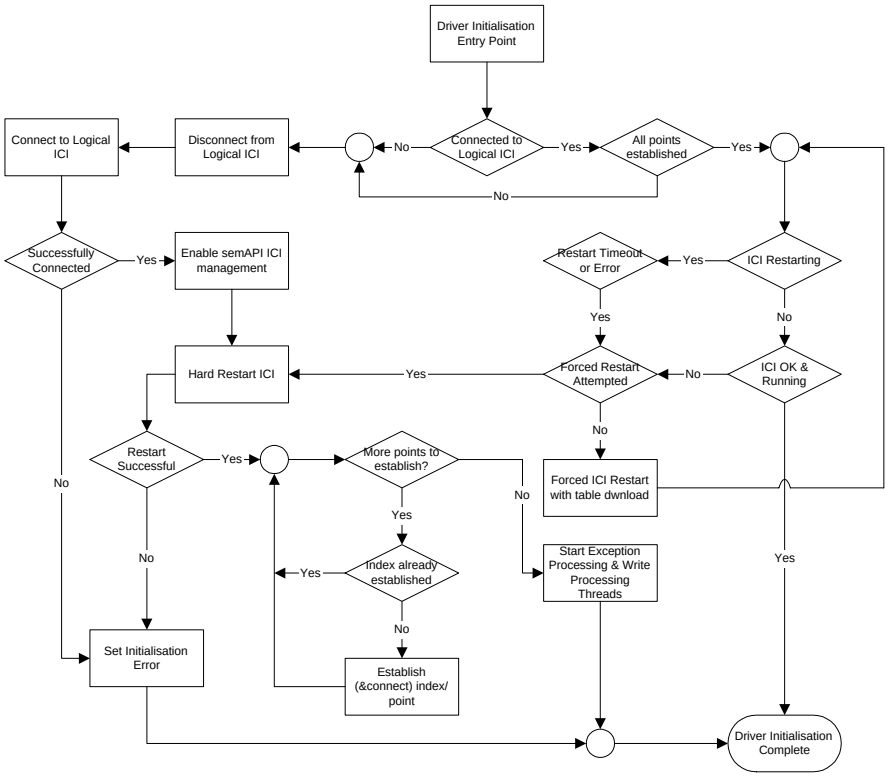


Figure 3.3-1 Driver initialisation flow diagram

3.3.3 Driver data exception handling

The driver data exception handling flow diagram is shown in Figure 3.3-2. The semAPI function to read exception reports from the ICI is a blocking function. As a result the data exception handling is performed in a separate thread to the main driver. This thread is created as a child thread of the main driver thread during unit initialisation and started by the initialisation thread when the ICI has been successfully initialised. Errors encountered while processing data exceptions are sent to CitectSCADA by raising an error in Read Dcb's.

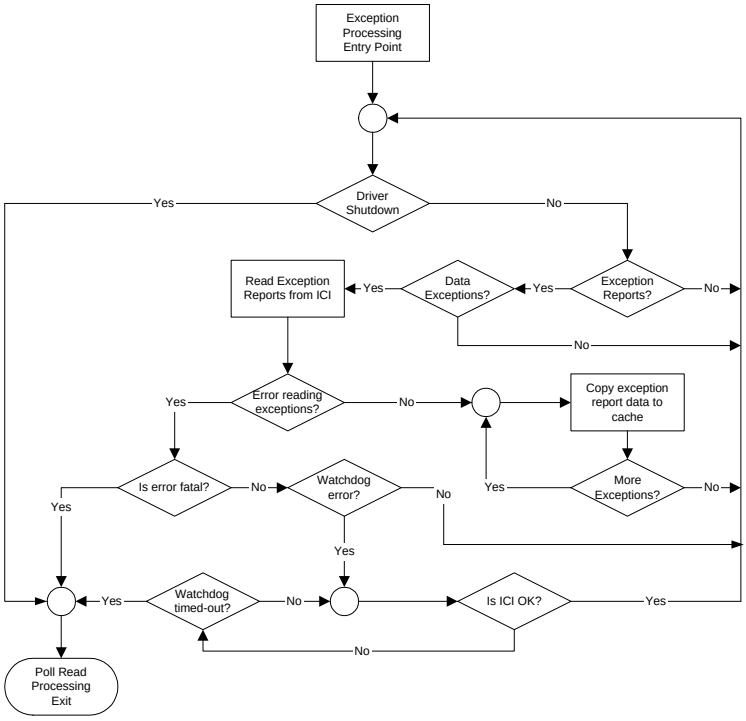


Figure 3.3-2 Driver Exception handling flow diagram

3.3.4 Driver poll data read processing

The driver data exception handling flow diagram is shown in Figure 3.3-3. The poll read processing is invoked by the read processing thread. Polled read processing is ratioed to exception processing, to ensure that the driver is not blocked by too many poll reads. The driver also limits the number of outstanding poll requests through the user parameter, MaxPendingPolls. The driver will only submit a poll request if there are fewer polls reads outstanding than the MaxPendingPolls parameter, and the last elapsed time between requests for the poll read item is greater than the user parameter, PolledDataPollTime (in milliseconds).

semAPI functions used to submit poll data requests (such as demand module status) are invoked using the quick function method. A quick message ID (see **Error! Reference source not found.**) is returned to the driver if the poll is submitted successfully, and can be used to check whether a reply has been received.

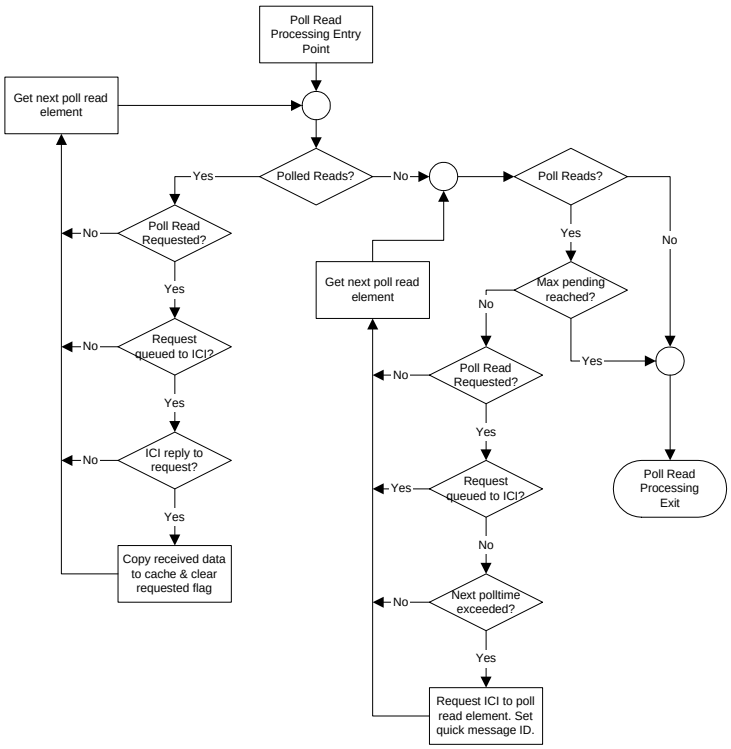


Figure 3.3-3 Driver Poll read processing flow diagram

3.3.5 Driver write handling

The driver CitectSCADA write Dcb handling flow diagram is shown in Figure 3.3-4. The semAPI function call to perform a control output point is blocking. As a result, the write-handling task is run in a separate thread to the main driver. This thread is created as a child thread of the main driver thread during unit initialisation, and started by the initialisation thread when the ICI has been successfully initialised. All CitectSCADA write requests for a unit are queued and serviced sequentially by this thread.

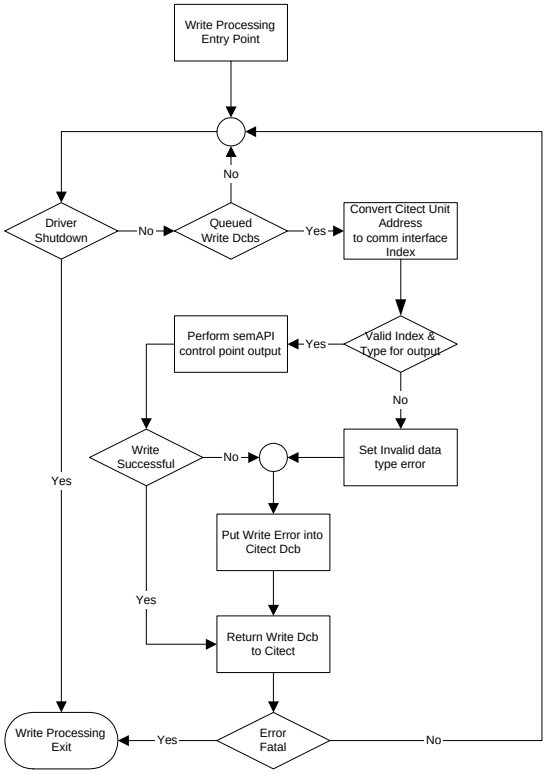


Figure 3.3-4 Driver CitectSCADA write-handling flow diagram

3.3.6 Driver read Processing

The driver CitectSCADA read Dcb handling flow diagram is shown in Figure 3.3-5. The tasks outlined only reference the memory-cache and do not involve any direct calls to semAPI. The read processing is performed in the main driver thread by the ReadDcb function. Errors that are encountered by the data exception processing thread are returned as an error in the next CitectSCADA Read Dcb.

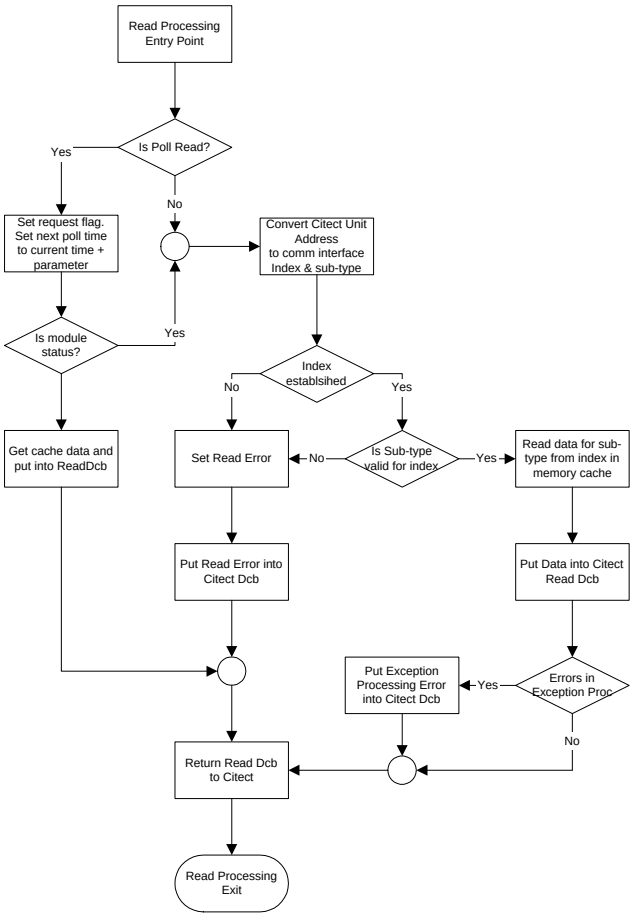


Figure 3.3-5 Driver CitectSCADA read-handling flow diagram

3.3.7 Multi-threading Protection

Thread/memory protection, using critical sections, is employed to ensure that the data exception handling thread does not access the memory-cache while read processing is in progress. This will ensure that memory read-write contention cannot occur and the data that is returned to CitectSCADA is consistent.

3.3.8 Driver flow examples

3.3.8.1 Unit initialisation for base variable

A CitectSCADA variable with address PV5,1,2,3,45 (PV Point Type, Index = 5, Ring = 1, Node = 2, Module = 3, Block = 45) needs to be established. During this phase the driver will refer to the _driver.rdb containing the INFI90 address within the CitectSCADA unit address and type.

The 5th memory cache element (index 5) will be updated to reflect that the index is being used, the Bailey point type is PV and the established flag will be set if the point was established successfully.

3.3.8.2 Unit initialisation for variable with sub-type

A CitectSCADA variable with address PV5,1,2,3,45.HL (PV Point Type, Index = 5, Ring = 1, Node = 2, Module = 3, Block = 45, Sub-Type = 6 (HL)) needs to be established.

The driver will see that the 'Used' flag has been set in the memory cache for index 5 (as per the example in 3.3.8.1) so the point does not need to be established again.

3.3.8.3 Exception report for communication interface index 5

When the back-end thread servicing incoming exceptions processes an exception report for communication index 5. The interface index is used as an index into the memory cache table and the exception reports value table and timestamp will be copied to the element's Data and timestamp fields.

3.3.8.4 CitectSCADA read request for real point

If CitectSCADA requests a read of a point with unit address 0x05, Raw Data Type RDT_REAL.
The unit address will be converted to index and sub-type (see 3.2 for Real data type conversion rule) resulting in a communication index of 5 and no sub-type. The value table of the 5th memory cache element is then accessed to get the floating-point value for the PV and returned to CitectSCADA.

3.3.8.5 CitectSCADA read request for digital point

If CitectSCADA requests a read of a point with unit address 0xA6, Raw Data Type RDT_DIGITAL.
The unit address will be converted to index and sub-type (see 3.2 for Real data type conversion rule) resulting in a communication index of 5 and a sub-type index of 6. The value table of the 5th memory cache element is then accessed to get the value of 6th sub-type for the PV (high limit alarm, HL) and returned to CitectSCADA.

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4. Driver implementation

The driver has been implemented using Microsoft Visual C++ 6.0, CitectSCADA for Windows v5.21 Service Pack C, Elsag Bailey semAPI Development Library 2.0 & semAPI Run-Time Library 2.1.

4.1 Development resources

4.1.1 Documents

- CitectSCADA Bailey Driver Specification (BaileySAF002.doc) modified on 8/12/1998 by Simon Frost.
- Bailey semAPI NT/95 Run-Time Interface (Version 2.1)
  - Introduction and Installation Manual.
- Bailey semAPI Developer Interface (Version 2.1)
  - Introduction and Installation Manual.
  - Introduction and User Manual.
- Bailey Net90 Enhanced CIU Programmer's Reference Manual.
- Bailey Infi90 Function Code Application Manual.
- Bailey Cnet-to-Computer Communications Interface (INICI03)

4.1.2 Software / Hardware

INICI03A INFI-NET to Host Computer Communication Interface.
semAPI Run-Time Version 2.1 (or later), hardware protection key and software licensing.
CitectSCADA v5.21 Service Pack C.

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5. Appendix B – Module dependant status'

The information returned by module status is dependent on the type of module. Each supported module is described below.
Status bits that are not included in each table have no meaning for that module.

5.1 NCTM01, IMAOM01, CTT01/02

These modules do not return any status information apart from mode and errors exist.

5.2 NPIM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	PCUs(nodes) offline on central/satelite
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Node's combined environment status flag
MSxxxxx.ST6	LIM internal error
MSxxxxx.ST7	Receive errors on channel 1
MSxxxxx.ST8	Receive errors on channel 2
MSxxxxx.ST9	Transmit errors on channel 1
MSxxxxx.ST10	Transmit errors on channel 2

5.3 NAMM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	Summary local input status

5.4 NCOM02, IMCOM03/04, IMQRC01, CLC01/02/03/04, NLCM01/02/03

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Summary remote input status
MSxxxxx.ST5	Summary local input status
MSxxxxx.ST6	Summary segment alarm status
MSxxxxx.ST7	Nonvolatile memory failure
MSxxxxx.ST8	Nonvolatile memory initialized
MSxxxxx.ST9	Digital station status

5.5 NLMM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Summary remote input status
MSxxxxx.ST6	Summary local input status
MSxxxxx.ST7	Nonvolatile memory initialized
MSxxxxx.ST8	Nonvloatile memory initialized
MSxxxxx.ST9	Digital station status

5.6 NLIM01/02, NBIM01, INBMI02

CitectSCADA Tag Format	Description
MSxxxxx.ST3	PCUs (nodes) offline on central/satellite
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Node's combined environment status flag
MSxxxxx.ST6	Backup module configured
MSxxxxx.ST7	Status of backup module
MSxxxxx.ST8	Primary/secondary indicator
MSxxxxx.ST9	Receive errors on channel 1
MSxxxxx.ST10	Receive errors on channel 2
MSxxxxx.ST11	Transmit errors on channel 1
MSxxxxx.ST12	Transmit errors on channel 2

5.7 NAMM02/IMAMM03

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Backup status
MSxxxxx.ST5	Summary local input status
MSxxxxx.ST6	Nonvolatile memory failure
MSxxxxx.ST7	Nonvolatile memory initialised
MSxxxxx.ST8	Master references out of range
MSxxxxx.ST9	Summary open TC status
MSxxxxx.ST10	Slave configuration problem
MSxxxxx.ST11	Summary calibration status
MSxxxxx.ST12	Summary cold junction RTD status
MSxxxxx.ST13	Summary of points not in service
MSxxxxx.ST14	Summary NVM power failure during write
MSxxxxx.ST15	Failed configuration NVM (NAMM02 only)
MSxxxxx.ST16	Failed calibration NVM (NAMM02 only)
MSxxxxx.ST17	Configuration NVM initialised (NAMM02 only)
MSxxxxx.ST18	Calibration NVM initialised (NAMM02 only)

5.8 INPTM01, NLSM01/INPCT01, CIC01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Summary remote input status
MSxxxxx.ST5	Summary local input status
MSxxxxx.ST6	PCUs off-line on central/satellite
MSxxxxx.ST7	Module memory overflow
MSxxxxx.ST8	Node's combined environment status flag
MSxxxxx.ST9	LIM internal error
MSxxxxx.ST10	Receive errors on channel 1
MSxxxxx.ST11	Receive errors on channel 2
MSxxxxx.ST12	Transmit errors on channel 1
MSxxxxx.ST13	Transmit errors on channel 2

5.9 NMFC01/02, IMMFC03/04/05, IMMPC01, CBC01, CSB01, SLC01/21

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Backup status
MSxxxxx.ST5	Summary remote input status
MSxxxxx.ST6	Summary local input status
MSxxxxx.ST7	Online configuration changes being made
MSxxxxx.ST8	Nonvolatile memory failure
MSxxxxx.ST9	Nonvolatile memory initialised
MSxxxxx.ST10	Digital station status

5.10 IMLMM02

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Summary remote input status
MSxxxxx.ST6	Summary local input status
MSxxxxx.ST7	Nonvolatile memory failure
MSxxxxx.ST8	Nonvolatile memory initialized
MSxxxxx.ST9	Digital station status

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5.11 INBTM01, NLIM01/02

CitectSCADA Tag Format	Description
MSxxxxx.ST3	Online, offline status
MSxxxxx.ST4	PCUs offline on central/satellite
MSxxxxx.ST5	Module memory overflow
MSxxxxx.ST6	LIM internal error
MSxxxxx.ST7	Receive errors on channel 1
MSxxxxx.ST8	Receive errors on channel 2
MSxxxxx.ST9	Transmit errors on channel 1
MSxxxxx.ST10	Transmit errors on channel 2

5.12 NGCM01/INPPT01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	Online, offline status
MSxxxxx.ST4	Backup status
MSxxxxx.ST5	Summary remote input status
MSxxxxx.ST6	Summary local input status
MSxxxxx.ST7	Nonvolatile memory failure
MSxxxxx.ST8	Nonvolatile memory initialised
MSxxxxx.ST9	PCUs offline on central/satellite
MSxxxxx.ST10	Node's combined environment status flag
MSxxxxx.ST11	Status of serial link one
MSxxxxx.ST12	Status of serial link two
MSxxxxx.ST13	LIM internal error
MSxxxxx.ST14	Receive errors on channel 1
MSxxxxx.ST15	Receive errors on channel 2
MSxxxxx.ST16	Transmit errors on channel 1
MSxxxxx.ST17	Transmit errors on channel 2

5.13 INIPT01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	I/O status for data sourced by satellite loops PCUs
MSxxxxx.ST4	I/O status for data sourced by central loops PCUs
MSxxxxx.ST5	Nonvolatile memory initialized
MSxxxxx.ST6	PCUs offline on central/satellite loop
MSxxxxx.ST7	Module memory overflow
MSxxxxx.ST8	Node's combined environment status flag
MSxxxxx.ST9	Backup module configured
MSxxxxx.ST10	Status of backup module
MSxxxxx.ST11	Offline bridge for remote loop
MSxxxxx.ST12	Node's environment status flag for central loop
MSxxxxx.ST13	Receive errors on channel 1 of central loop
MSxxxxx.ST14	Receive errors on channel 2 of central loop
MSxxxxx.ST15	Transmit errors on channel 1 of central loop
MSxxxxx.ST16	Transmit errors on channel 2 of central loop
MSxxxxx.ST17	Ring communication failure for central loop
MSxxxxx.ST18	Node's environment status flag for satellite loop
MSxxxxx.ST19	Receive errors on channel 1 of satellite loop
MSxxxxx.ST20	Receive errors on channel 2 of satellite loop
MSxxxxx.ST21	Transmit errors on channel 1 of satellite loop
MSxxxxx.ST22	Transmit errors on channel 2 of satellite loop
MSxxxxx.ST23	Ring communication failure for satellite loop
MSxxxxx.ST24	NIS loop relay drive transistor 1 failure on central loop
MSxxxxx.ST25	NIS loop relay drive transistor 2 failure on central loop
MSxxxxx.ST26	NIS loop relay drive transistor 1 failure on satellite loop
MSxxxxx.ST27	NIS loop relay drive transistor 2 failure on satellite loop
MSxxxxx.ST28	NIS receive channel 1 disable on central loop
MSxxxxx.ST29	NIS receive channel 2 disable on central loop
MSxxxxx.ST30	NIS receive channel 1 disable on satellite loop
MSxxxxx.ST31	NIS receive channel 2 disable on satellite loop

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5.14 INIPT02

CitectSCADA Tag Format	Description
MSxxxxx.ST3	I/O status for data sourced by satellite loops PCUs
MSxxxxx.ST4	I/O status for data sourced by central loops PCUs
MSxxxxx.ST5	Nonvolatile memory initialized
MSxxxxx.ST6	PCUs offline on central/satellite loop
MSxxxxx.ST7	Module memory overflow
MSxxxxx.ST8	Serial link composite status
MSxxxxx.ST9	Status of backup module
MSxxxxx.ST10	Offline bridge for remote loop
MSxxxxx.ST11	Primary/secondary indicator
MSxxxxx.ST12	Node's combined environment status flag
MSxxxxx.ST13	Receive errors on channel 1
MSxxxxx.ST14	Receive errors on channel 2
MSxxxxx.ST15	Transmit errors on channel 1
MSxxxxx.ST16	Transmit errors on channel 2
MSxxxxx.ST17	Receiver idle on channel 1
MSxxxxx.ST18	Receiver idle on channel 2
MSxxxxx.ST19	Ring communication failure for local loop
MSxxxxx.ST20	Staus of serial link 1
MSxxxxx.ST21	NIS loop relay drive transistor 1 failure
MSxxxxx.ST22	NIS loop relay drive transistor 2 failure
MSxxxxx.ST23	NIS receive channel 1 disable
MSxxxxx.ST24	NIS receive channel 2 disable

5.15 ICI Modules: INICT01/03, IIMCP01/02

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Summary remote input status
MSxxxxx.ST5	Summary local input status
MSxxxxx.ST6	Module in configure mode
MSxxxxx.ST7	Module in error mode
MSxxxxx.ST8	PCUs offline on central/satellite loop
MSxxxxx.ST9	Module memory overflow
MSxxxxx.ST10	Offline bridge for remote loop
MSxxxxx.ST11	Security device failure
MSxxxxx.ST12	Node's combined environment status flag
MSxxxxx.ST13	Receive errors on channel 1
MSxxxxx.ST14	Receive errors on channel 2
MSxxxxx.ST15	Transmit errors on channel 1
MSxxxxx.ST16	Transmit errors on channel 2
MSxxxxx.ST17	Receiver idle on channel 1
MSxxxxx.ST18	Receiver idle on channel 2
MSxxxxx.ST19	Ring communication failure for local loop
MSxxxxx.ST20	NIS loop relay drive transistor 1 failure
MSxxxxx.ST21	NIS loop relay drive transistor 2 failure
MSxxxxx.ST22	NIS receive channel 1 disable
MSxxxxx.ST23	NIS receive channel 2 disable

5.16 NSBM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	PCUs offline on central/satellite loop
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Backup module configured
MSxxxxx.ST6	Status of backup module
MSxxxxx.ST7	Primary/secondary indicator
MSxxxxx.ST8	Node's combined environment status flag
MSxxxxx.ST9	Receive errors on channel 1
MSxxxxx.ST10	Receive errors on channel 2
MSxxxxx.ST11	Transmit errors on channel 1
MSxxxxx.ST12	Transmit errors on channel 2
MSxxxxx.ST13	Receiver idle on channel 1
MSxxxxx.ST14	Receiver idle on channel 2
MSxxxxx.ST15	Ring communication failure for local loop
MSxxxxx.ST16	NIS loop relay drive transistor 1 failure
MSxxxxx.ST17	NIS loop relay drive transistor 2 failure
MSxxxxx.ST18	NIS receive channel 1 disable
MSxxxxx.ST19	NIS receive channel 2 disable

5.17 NSCM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	Backup status
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	NVM full
MSxxxxx.ST6	Nonvolatile memory failure
MSxxxxx.ST7	Nonvolatile memory initialised
MSxxxxx.ST8	Node's combined environment status flag
MSxxxxx.ST9	Receive errors on channel 1
MSxxxxx.ST10	Receive errors on channel 2
MSxxxxx.ST11	Transmit errors on channel 1
MSxxxxx.ST12	Transmit errors on channel 2
MSxxxxx.ST13	Receiver idle on channel 1
MSxxxxx.ST14	Receiver idle on channel 2
MSxxxxx.ST15	Port 0 linked
MSxxxxx.ST16	Port 0 mode
MSxxxxx.ST17	Port 1 linked
MSxxxxx.ST18	Port 1 mode
MSxxxxx.ST19	Port 2 linked
MSxxxxx.ST20	Port 2 mode
MSxxxxx.ST21	Port 3 linked
MSxxxxx.ST22	Port 3 mode
MSxxxxx.ST23	Port 4 linked
MSxxxxx.ST24	Port 4 mode
MSxxxxx.ST25	Port 5 linked
MSxxxxx.ST26	Port 5 mode
MSxxxxx.ST27	Port 6 linked
MSxxxxx.ST28	Port 6 mode
MSxxxxx.ST29	Port 7 linked
MSxxxxx.ST30	Port 7 mode

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5.18 INIIT01/03

CitectSCADA Tag Format	Description
MSxxxxx.ST3	I/O status for data sourced by satellite loops PCUs
MSxxxxx.ST4	I/O status for data sourced by central loops PCUs
MSxxxxx.ST5	PCUs offline on central/satellite loop
MSxxxxx.ST6	Module memory overflow
MSxxxxx.ST7	Backup module configured
MSxxxxx.ST8	Status of backup module
MSxxxxx.ST9	Primary/secondary indicator
MSxxxxx.ST10	Node's environment status flag for central loop
MSxxxxx.ST11	Receive errors on channel 1 of central loop
MSxxxxx.ST12	Receive errors on channel 2 of central loop
MSxxxxx.ST13	Transmit errors on channel 1 of central loop
MSxxxxx.ST14	Transmit errors on channel 2 of central loop
MSxxxxx.ST15	Ring communication failure for central loop
MSxxxxx.ST16	Node's environment staus flag for satellite loop
MSxxxxx.ST17	Receive errors on channel 1 of satellite loop
MSxxxxx.ST18	Receive errors on channel 2 of satellite loop
MSxxxxx.ST19	Transmit errors on channel 1 of satellite loop
MSxxxxx.ST20	Transmit errors on channel 2 of satellite loop
MSxxxxx.ST21	Ring communication failure for satellite loop
MSxxxxx.ST22	NIS loop relay drive transistor 1 failure on central loop
MSxxxxx.ST23	NIS loop relay drive transistor 2 failure on central loop
MSxxxxx.ST24	NIS loop relay drive transistor 1 failure on satellite loop
MSxxxxx.ST25	NIS loop relay drive transistor 2 failure on satellite loop
MSxxxxx.ST26	NIS receive channel 1 disable on central loop
MSxxxxx.ST27	NIS receive channel 2 disable on central loop
MSxxxxx.ST28	NIS receive channel 1 disable on satellite channel
MSxxxxx.ST29	NIS receive channel 2 disable on satellite channel

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5.19 MFP Modules: IMMFP01/02/03/03B, BRC-100

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Backup status
MSxxxxx.ST5	Summary remote input status
MSxxxxx.ST6	Summary local input status
MSxxxxx.ST7	Online configuration changes being made
MSxxxxx.ST8	Nonvolatile memory failure
MSxxxxx.ST9	Nonvolatile memory initialised
MSxxxxx.ST10	Digital station status
MSxxxxx.ST11	Controlway channel A status
MSxxxxx.ST12	Controlway channel B status
MSxxxxx.ST13	Redundancy link channel 1 failure
MSxxxxx.ST14	Redundancy link channel 2 failure
MSxxxxx.ST15	Peripheral failure
MSxxxxx.ST16	Harmony net channel A failure
MSxxxxx.ST17	Harmony net channel B failure
MSxxxxx.ST18	Simulation enabled
MSxxxxx.ST19	Simulation running
MSxxxxx.ST20	Simulation time rate
MSxxxxx.ST21	Harmony net channel A relay fault
MSxxxxx.ST22	Harmony net channel B relay fault
MSxxxxx.ST23	Module rank
MSxxxxx.ST24	Configuration complete
MSxxxxx.ST25	Checkpoint complete
MSxxxxx.ST26	State of redundancy ID dipswitch pole
MSxxxxx.ST27	Redundant module expected (latched indication)
MSxxxxx.ST28	State of online configuration dipswitch pole
MSxxxxx.ST29	Redundancy detected
MSxxxxx.ST30	Redundancy nvm overrun (latched indication)

5.20 INNPM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	PCUs offline on central/satellite loop
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Backup module configured
MSxxxxx.ST6	Status of backup module
MSxxxxx.ST7	Primary/secondary indicator
MSxxxxx.ST8	Node's combined environment status flag
MSxxxxx.ST9	Receive errors on channel 1
MSxxxxx.ST10	Receive errors on channel 2
MSxxxxx.ST11	Transmit errors on channel 1
MSxxxxx.ST12	Transmit errors on channel 2
MSxxxxx.ST13	Receiver idle on channel 1
MSxxxxx.ST14	Receiver idle on channel 2
MSxxxxx.ST15	Ring communication failure for local loop
MSxxxxx.ST16	Controlway channel A status
MSxxxxx.ST17	Controlway channel B status
MSxxxxx.ST18	NIS loop relay drive transistor 1 failure
MSxxxxx.ST19	NIS loop relay drive transistor 2 failure
MSxxxxx.ST20	NIS receive channel 1 disable

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5.21 INIIT02

CitectSCADA Tag Format	Description
MSxxxxx.ST3	I/O status for data sourced by satellite loops PCUs
MSxxxxx.ST4	I/O status for data sourced by central loops PCUs
MSxxxxx.ST5	PCUs offline on central/satellite loop
MSxxxxx.ST6	Module memory overflow
MSxxxxx.ST7	Serial link composite status
MSxxxxx.ST8	Backup module configured
MSxxxxx.ST9	Status of backup module
MSxxxxx.ST10	Offline bridge for remote loop
MSxxxxx.ST11	Primary/secondary indicator
MSxxxxx.ST12	Node's combined environment status flag
MSxxxxx.ST13	Receive errors on channel 1
MSxxxxx.ST14	Receiver errors on channel 2
MSxxxxx.ST15	Transmit errors on channel 1
MSxxxxx.ST16	Transmit errors on channel 2
MSxxxxx.ST17	Receiver idle on channel 1
MSxxxxx.ST18	Receiver idle on channel 2
MSxxxxx.ST19	Ring communication failure for loop
MSxxxxx.ST20	Status of serial link 1
MSxxxxx.ST21	Digital output #3 of slave module is in energised state
MSxxxxx.ST22	Status of serial link 2
MSxxxxx.ST23	NIS loop relay drive transistor 1 failure
MSxxxxx.ST24	NIS loop relay drive transistor 2 failure
MSxxxxx.ST25	NIS receive channel 1 disable
MSxxxxx.ST26	NIS receive channel 2 disable

5.22 INCPM02/03

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Summary remote input status
MSxxxxx.ST5	Summary local input status
MSxxxxx.ST6	PCUs offline
MSxxxxx.ST7	Module memory overflow
MSxxxxx.ST8	Offline bridge for remote loop
MSxxxxx.ST9	Security device failure
MSxxxxx.ST10	Controllway channel A status
MSxxxxx.ST11	Controllway channel B status
MSxxxxx.ST12	NIS loop relay drive transistor 1 failure
MSxxxxx.ST13	NIS loop relay drive transistor 2 failure
MSxxxxx.ST14	NIS receive channel 1 disable
MSxxxxx.ST15	NIS receive channel 2 disable

5.23 INSEM01

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Summary remote input status
MSxxxxx.ST5	Summary local input status
MSxxxxx.ST6	Module in configure mode
MSxxxxx.ST7	Module in error mode
MSxxxxx.ST8	PCUs offline on central/satellite loop
MSxxxxx.ST9	Module memory overflow
MSxxxxx.ST10	Offline bridge for remote loop
MSxxxxx.ST11	Node's combined environment status flag
MSxxxxx.ST12	Receive errors on channel 1
MSxxxxx.ST13	Receive errors on channel 2
MSxxxxx.ST14	Transmit errors on channel 1
MSxxxxx.ST15	Transmit errors on channel 2
MSxxxxx.ST16	Receiver idle on channel 1
MSxxxxx.ST17	Receiver idle on channel 2
MSxxxxx.ST18	Ring communication failure for local loop

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5.24 Harmony processor – Controller status

CitectSCADA Tag Format	Description
MSxxxxx.ST3	First time in execute
MSxxxxx.ST4	Redundant failure status
MSxxxxx.ST5	Summary remote input status
MSxxxxx.ST6	Summary local input status
MSxxxxx.ST7	Nonvolatile memory failure
MSxxxxx.ST8	Nonvolatile memory initialised
MSxxxxx.ST9	Station error status
MSxxxxx.ST10	Controller Harmony net A failure
MSxxxxx.ST11	Controller Harmony net B failure
MSxxxxx.ST12	Redundancy link channel 1 failure
MSxxxxx.ST13	Redundancy link channel 2 failure
MSxxxxx.ST14	Peripheral failure
MSxxxxx.ST15	D-Bus communication established
MSxxxxx.ST16	I/O Harmony net channel A failure
MSxxxxx.ST17	I/O Harmony net channel B failure
MSxxxxx.ST18	Module memory overflow
MSxxxxx.ST19	NIS channel 1 failure
MSxxxxx.ST20	NIS channel 2 failure
MSxxxxx.ST21	Logic power input A status
MSxxxxx.ST22	Logic power input B status
MSxxxxx.ST23	Status output alarm
MSxxxxx.ST24	Redundancy nvm overrun (latched indication)
MSxxxxx.ST25	NIS offline
MSxxxxx.ST26	PCU offline
MSxxxxx.ST27	Receive errors on channel 1
MSxxxxx.ST28	Receive errors on channel 2
MSxxxxx.ST29	Transmit errors on channel 1
MSxxxxx.ST30	Transmit errors on channel 2
MSxxxxx.ST31	Ring communication failure for local loop

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5.25 Harmony processor – NPM status

CitectSCADA Tag Format	Description
MSxxxxx.ST3	PCUs offline on central/satellite loop
MSxxxxx.ST4	Module memory overflow
MSxxxxx.ST5	Redundant module expected
MSxxxxx.ST6	Redundant failure status
MSxxxxx.ST7	Primary/secondary indicator
MSxxxxx.ST8	Node's combined environment status flag
MSxxxxx.ST9	Receive errors on channel 1
MSxxxxx.ST10	Receive errors on channel 2
MSxxxxx.ST11	Transmit errors on channel 1
MSxxxxx.ST12	Transmit errors on channel 2
MSxxxxx.ST13	Receiver idle on channel 1
MSxxxxx.ST14	Receiver idle on channel 2
MSxxxxx.ST15	Ring communication failure for local loop
MSxxxxx.ST16	Controller Harmony net A status
MSxxxxx.ST17	Controller Harmony net B status
MSxxxxx.ST18	NIS loop relay drive transistor 1 failure
MSxxxxx.ST19	NIS loop relay drive transistor 2 failure
MSxxxxx.ST20	NIS receive channel 1 disable
MSxxxxx.ST21	NIS receive channel 2 disable

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